

Book III

Persistence, Transport, and Observational Transfer

Elements of Nested Fibrational Cosmology

Contents

III.0	Purpose	4
1.	Imports and Standing Rules	4
2.	Persistent Observables and Transfer Maps	5
3.	Persistence Lemmas	6
4.	Persistence and Compatibility Propositions	7
5.	The Interface-Relative Stability Functional	8
6.	Finite Balance and the Unified Coercive-Transport Inequality	9
7.	Transfer Coercivity	11
8.	Observational-Transfer Objects and POT Preparation	12
9.	Governing Theorem of Book III	13
10.	Conserved Charges and Cheeger Audit Infrastructure	15
11.	Variable Recovery: The Dynamical Core of the Ladder	17
12.	Synchronization of Recovered Variables: One Obstruction, Supported on Cycle Rank	49
13.	β_1 Cycle Rank as Canonical Complexity Parameter	51
14.	Boundary of Book III	52
15.	Handoff to Book IV	53
16.	Dependency Ledger	54

III.0 Purpose

Book III is the motion and comparison book of the canonical spine.

Book I established that admissible testing produces a coherent observational quotient and that every finite refinement chain stabilizes. Book II established that, under LS-2, stabilized local observable outcomes are determined by bounded collar data and that finite collar alphabets impose hard multiplicity bounds. Book III now asks: how does certified local structure *persist* and *transfer* across admissible refinements and contexts?

This question is essential because Book IV must certify the existence of a universal source object U that all lawful downstream structures factor through. Factorization requires that certified observables from different local regimes can be coherently compared. Coherent comparison requires a transport law. Book III provides that law.

The central theorem of the book is the Unified Coercive-Transport Inequality: a branch-agnostic bound on how the interface-relative stability functional changes under admissible enlargement. This inequality encodes the finite balance between transported bulk stability, internal defect, and boundary defect. Its proof uses only the defect ledger of Book I and the LS-2 boundary law of Book II.

Book III is explicitly branch-agnostic. It does not specialize to Yang–Mills, Navier–Stokes, gravity, or any other specific endpoint program. All such specializations belong in derived branch books. What this book establishes is the upstream source law from which those specializations descend.

1. Imports and Standing Rules

Imports from Book I. Finite configuration spaces Ω , admissible test families $\mathcal{T}_n$, observational equivalence $\sim_n$, quotients Σ_n , stabilized quotient Σ_∞ , survivor chains, one-step defect $\delta(C_n)$, defect ledger $\Delta(C_n \rightsquigarrow C_m)$, and the Observational Quotient Theorem.

Imports from Book II. Regions U , collars $\partial_r U$, stabilized collar types, the boundary law map Φ_U under LS-2, realized stabilized outcomes $O(U)$, boundary capacity $\text{Cap}_\partial(U)$, and the Boundary Stabilization Theorem.

STANDING RULE 1.1 ([D]). No persistent observable may depend on presentation-level distinctions that were eliminated by the observational quotient of Book I. Persistence is quotient-visible.

STANDING RULE 1.2 ([D]). No admissible transfer may destroy the stabilized local structure certified in Book II without recording a transport obstruction. Transfer must respect stabilization.

STANDING RULE 1.3 ([D]). Claims about gain, loss, or conservation of observable content must descend from certified finite balance data. Balance before interpretation.

STANDING RULE 1.4 ([D]). Observable comparison across contexts has canonical force only through explicitly declared transfer maps.

2. Persistent Observables and Transfer Maps

DEFINITION 2.1 ([D]). A *persistent observable* is a certified observable whose value is preserved or lawfully transformed across the declared class of admissible refinements or comparison contexts. Formally, it is a function $\mathcal{O}$ defined on stabilized quotient classes that satisfies the compatibility condition declared for the relevant class of transfer maps.

DEFINITION 2.2 ([D]). A *transfer map* $f : (U, \Sigma_\infty) \rightarrow (U', \Sigma'_\infty)$ is a certified witness-preserving comparison map carrying admissible observable data from one context or refinement stage to another. It must be compatible with observational quotient structure: if $\omega \sim_\infty \omega'$ in the source context, then $f(\omega) \sim'_\infty f(\omega')$ in the target context.

DEFINITION 2.3 ([D]). A *transport class* is an equivalence class of admissible observable data under lawful transfer: two observations are in the same transport class if one is the transfer-image of the other via a declared transfer map.

DEFINITION 2.4 ([D]). A *compatible realization* is a realization whose certified observable content agrees with the required transfer and persistence conditions.

DEFINITION 2.5 ([D]). *Observational drift* is the certified change in an observable under admissible transfer, prior to identifying whether that change is lawful, bounded, or obstructive.

DEFINITION 2.6 ([D]). *Transfer compatibility* is the condition that a family of observables behaves coherently under the declared class of transfer maps: the transport-class structure is preserved under composition of transfer maps.

DEFINITION 2.7 ([D]). A *transport obstruction* is a certified failure of persistence, compatibility, or lawful transfer. It records the exact stage and context at which a persistent observable fails to transfer lawfully.

DEFINITION 2.8 ([D]). An *observational-transfer object* is an admissible quotient-visible object equipped with the persistence and transport structure needed for lawful comparison across contexts: a stabilized observational quotient, a family of declared transfer maps, and a defect-ledger encoding observational drift along admissible transfer chains.

DEFINITION 2.9 ([D]). A *composable transfer chain* is a finite or admissibly indexed sequence of transfer maps

$$f_0 : (U_0, \Sigma_\infty^{(0)}) \rightarrow (U_1, \Sigma_\infty^{(1)}), \quad f_1 : (U_1, \Sigma_\infty^{(1)}) \rightarrow (U_2, \Sigma_\infty^{(2)}), \quad \dots$$

for which each successive transfer is lawful and the composite $f_n \circ \dots \circ f_0$ is well-defined on the relevant certified observable content.

DEFINITION 2.10 ([D]). *Transfer coercivity* is a certified constraint limiting uncontrolled observational drift along admissible transfer chains. It asserts that the accumulated change in a declared observable family across a composable transfer chain is bounded by a quantity determined by the declared structural data.

REMARK 2.11 ([R]). Transfer coercivity is the bridge concept that will allow Book IV to assemble local fiber data into a global persistent source object. The coercive bound is what prevents the persistence diagram from diverging as one takes larger and larger regimes: it controls how much observational content can be “lost” at each transfer step.

3. Persistence Lemmas

LEMMA 3.1 ([U]). *Persistent observables are invariant under replacement by quotient-equivalent representatives.*

PROOF. By Standing Rule 1.1 and the definition of persistent observable: a persistent observable’s value is determined by the stabilized quotient class, not by the choice of representative within that class. Two quotient-equivalent representatives belong to the same class, so the observable assigns them the same value. $\square$

LEMMA 3.2 ([U]). *Transfer maps preserve certified local compatibility whenever their domain and codomain satisfy the declared admissibility conditions.*

PROOF. By Definition 2.2, transfer maps are witness-preserving and compatible with quotient structure. Therefore any certified local compatibility relation in the source context — meaning any admissible relation detectable by the licensed test family — is preserved under the transfer map in the target context. $\square$

LEMMA 3.3 ([U]). *Transfer compatibility is invariant under lawful refinement of comparison stage.*

PROOF. Suppose a family of observables is transfer-compatible at comparison stage n . Under lawful refinement to stage $n + 1$, the quotient Σ_{n+1} refines Σ_n , and transfer maps compatible at stage n remain compatible at stage $n + 1$ because the finer partition only introduces additional distinctions that are also quotient-visible. Transfer compatibility, being a condition on equivalence classes, is preserved under refinement. $\square$

LEMMA 3.4 ([U]). *Observational drift is well-defined on certified observable families under declared transfer maps.*

PROOF. Observational drift measures the change $\mathcal{O}(f(\omega)) - \mathcal{O}(\omega)$ under a transfer map f . Since f is well-defined on quotient classes (Definition 2.2) and $\mathcal{O}$ is a function on quotient classes, the drift is well-defined. $\square$

LEMMA 3.5 ([U]). *Transport obstructions are quotient-visible and independent of surplus presentation structure.*

PROOF. A transport obstruction is defined as a certified failure of persistence, compatibility, or lawful transfer. Each of these conditions is formulated on quotient classes (by Definitions 2.1, 2.6, 2.2). Therefore the presence or absence of an obstruction is determined at the quotient level; presentation-level surplus cannot introduce or remove a genuine transport obstruction. $\square$

LEMMA 3.6 ([U]). *Composition of lawful transfer maps is well-defined on compatible realizations.*

PROOF. If f and g are lawful transfer maps with $\text{codomain}(f) = \text{domain}(g)$, then $g \circ f$ is a witness-preserving map compatible with quotient structure by the definition of transfer maps and the composability condition of Definition 2.9. On compatible realizations, the composition preserves the relevant certified observable content at each step. $\square$

4. Persistence and Compatibility Propositions

PROPOSITION 4.1 ([U]). *Persistent observables are well-defined on the admissible comparison arena.*

PROOF. By Lemma 3.1, persistent observables are quotient-visible. By the Observational Quotient Theorem (Book I, Theorem I.6.1), the quotient Σ_∞ is well-defined. Therefore persistent observables are well-defined functions on Σ_∞ . $\square$

PROPOSITION 4.2 ([U]). *Transfer compatibility is preserved under lawful refinement.*

PROOF. Lemma 3.3. $\square$

PROPOSITION 4.3 ([U]). *Compatible realizations induce the same transport class when related by lawful transfer.*

PROOF. If two realizations ω and ω' are related by a lawful transfer map f — meaning $f(\omega) = \omega'$ as quotient-class data — then they are in the same transport class by Definition 2.3. Transfer compatibility ensures this relation is well-defined and extends to the full observable family. $\square$

PROPOSITION 4.4 ([U]). *A transport obstruction prevents identification of the affected observable family as persistent.*

PROOF. A transport obstruction records a certified failure of persistence, compatibility, or lawful transfer (Definition 2.7). A persistent observable must be preserved or lawfully transformed under all declared transfer maps (Definition 2.1). An obstruction certifies that at least one transfer map fails this requirement; hence the affected observable family is not persistent in the declared context. $\square$

PROPOSITION 4.5 ([U]). *Transfer-compatible observable families define a lawful comparison structure across admissible contexts.*

PROOF. Composability of lawful transfer maps (Lemma 3.6) and transfer compatibility (Definition 2.6) together imply that the collection of transfer-compatible observable families, equipped with transfer maps, forms a directed comparison structure: identity transfers are lawful, transfer maps compose, and the transport-class structure is well-defined. $\square$

COROLLARY 4.6 ([U]). *Only transfer-compatible observable families may contribute to the source completion of Book IV.*

PROOF. The source completion in Book IV will assemble fiber data from all admissible regimes into a universal object. Assembly requires that data from different

regimes can be coherently compared, which requires transfer compatibility. Observable families with transport obstructions cannot be coherently assembled, hence they are excluded from the source-completion program. $\square$

5. The Interface-Relative Stability Functional

We now introduce the principal quantitative object of Book III. This functional organizes the balance between bulk stability and boundary burden in a form that is branch-agnostic: it applies to any admissible regime before specialization to any particular endpoint program.

DEFINITION 5.1 ([D]). For a region U_n at comparison stage n , the *bulk stability measure* $B_n^{\text{bulk}}(U_n)$ is the certified coercive content of U_n that is controlled by the Phase-A defect-floor architecture: the net certified observable content on U_n that resists further defect accumulation.

DEFINITION 5.2 ([D]). The *interface burden* $I_n(U_n)$ at stage n is the defect contribution at the boundary $\partial_{\rho^*}U_n$, determined by the LS-2 boundary law: the total certified defect load arising from the collar structure of U_n .

DEFINITION 5.3 ([D]). The *interface-relative stability functional* at stage n is

$$C_n(U_n) := B_n^{\text{bulk}}(U_n) - I_n(U_n).$$

REMARK 5.4 ([R]). The interface-relative stability functional C_n is defined at the pre-branch layer, before any specialization to Yang–Mills, Navier–Stokes, gravity, or any other specific endpoint. Its two components are both already present in the certified upstream machinery: B_n^{bulk} from the defect-floor structure of Book I, and I_n from the LS-2 boundary law of Book II. This universality is the source of the functional’s load-bearing character.

DEFINITION 5.5 ([D]). The *transport factor* at stage n , denoted $\Theta_n(U_n) \in [0, 1]$, is the certified multiplier expressing what fraction of $C_n(U_n)$ survives transport from U_n to U_{n+1} under the admissible enlargement.

DEFINITION 5.6 ([D]). The *transported internal defect* $E_n(U_n)$ is the certified defect increment arising from interior sites of U_n that survive enlargement but are not fully transported: the ledger increments $\Delta\Lambda$ at interior sites during the step $U_n \rightarrow U_{n+1}$.

DEFINITION 5.7 ([D]). The *interface/boundary defect* $B_n(U_n)$ is the certified defect increment arising at ∂U_n forced by the LS-2 boundary-determinacy structure: collar-type proliferation at the interface of U_n during the step $U_n \rightarrow U_{n+1}$.

REMARK 5.8 ([R]). Definitions 5.5–5.7 identify the three sources of observable change across one admissible enlargement step: transported bulk content (discounted by Θ_n), internal defect (E_n), and boundary defect (B_n). The Unified Coercive-Transport Inequality bounds C_{n+1} in terms of these three sources. No other sources are present: the defect ledger of Book I and the boundary law of Book II account for all of them.

6. Finite Balance and the Unified Coercive-Transport Inequality

DEFINITION 6.1 ([D]). A *finite balance datum* is the certified accounting relation describing the transport, gain, loss, or redistribution of observable content across one admissible enlargement step. For the interface-relative stability functional, it takes the form

$$C_{n+1}(U_{n+1}) \leq \Theta_n(U_n) C_n(U_n) + E_n(U_n) + B_n(U_n).$$

THEOREM 6.2 ([U]). (Unified Coercive-Transport Inequality.) *For every admissible enlargement $U_n \subseteq U_{n+1}$, the interface-relative stability functional satisfies*

$$C_{n+1}(U_{n+1}) \leq \Theta_n(U_n) C_n(U_n) + E_n(U_n) + B_n(U_n),$$

where $\Theta_n(U_n) \in [0, 1]$ is the transport factor, $E_n(U_n)$ is the transported internal defect, and $B_n(U_n)$ is the interface boundary defect.

PROOF. The bulk stability $B_{n+1}^{\text{bulk}}(U_{n+1})$ of the enlarged regime receives contributions from three distinct sources, which we track using the defect ledger of Book I and the LS-2 boundary law of Book II.

Source 1: Transported bulk. The bulk stability that survives transport from U_n to U_{n+1} is discounted by the transport factor $\Theta_n \in [0, 1]$, giving a contribution of at most $\Theta_n C_n(U_n)$. The transport factor accounts for the fact that not all of C_n survives enlargement: some certified content is absorbed by the expansion of the comparison regime.

Source 2: Internal defect E_n . Certified events in the interior of U_n that are not fully transported contribute defect ledger increments $\Delta\Lambda$ at interior sites during the enlargement step. These sum to $E_n(U_n)$, which is additive by the Ledger Additivity Proposition (Book I, Proposition I.5.4).

Source 3: Boundary defect B_n . Certified events at ∂U_n arise from the LS-2 boundary-determinacy structure (Book II, Theorem II.4.1): collar-type proliferation at the interface forces defect contributions $B_n(U_n) = \Delta_{\text{bdry}}(U_n)$. These are determined by the boundary law map Φ_{U_n} .

By additivity of the defect ledger across all three sources,

$$C_{n+1}(U_{n+1}) \leq \Theta_n C_n(U_n) + E_n(U_n) + B_n(U_n). \quad \square$$

REMARK 6.3 ([R]). The Unified Coercive-Transport Inequality is proved here in fully branch-agnostic form, depending only on the defect ledger of Book I and the LS-2 boundary law of Book II. It is a source-level theorem, upstream of all specializations. The three branch programs — Yang–Mills, Navier–Stokes, gravity — each specialize this inequality by fixing specific forms of Θ_n , E_n , and B_n appropriate to their endpoint targets. Those specializations belong in derived branch books, not in this book. Any citation of this inequality in a branch book must carry an explicit declaration of which specialization is being applied.

COROLLARY 6.4 ([U]). (Terminal-Observable Projection Theorem.) *The interface-relative stability functional C_n admits exactly two canonically distinct functorial projections to terminal observables:*

- (i) **Spectral projection** $\pi_{\text{YM}}: C_n \rightarrow g_\infty := \lambda_1(\Delta_A)$: the covariant Laplacian spectral gap. Closure condition: $g_\infty > 0$.
- (ii) **Decay projection** $\pi_{\text{NS}}: C_n \rightarrow \sup_n Q_n$: the supremum of the obstruction quotient. Closure condition: $\sup_n Q_n < \infty$.

Both projections are functorial with respect to the transition maps $\alpha_{R \rightarrow R'}$ of the POT category (Book IV). The two Millennium Prize problems addressed in the YM and NS branch books are the closure conditions of these two projections applied to the same upstream object C_n .

PROOF. Both projections are defined by specializing the decomposition $C_n = B_{\text{bulk}}(U_n) - I_n$ (Definition 5.3). π_{YM} takes B_{bulk} via the covariant Laplacian eigenvalue and I_n via the gap margin; functoriality follows from the transition map coherence of Theorem 6.5. π_{NS} takes C_n via the covering-time-normalized defect floor and Q_n via the quadratic rescaling; functoriality follows from Phase-A exhaustion enlargement compatibility (Theorem 6.6). The projections are distinct because their target observables are dimensionally incomparable: g_∞ is a spectral quantity (units of energy) while $\sup_n Q_n$ is a dimensionless ratio. $\square$

THEOREM 6.5 ([U]). *Finite balance data compose along composable transfer chains: if $U_n \rightarrow U_{n+1} \rightarrow \cdots \rightarrow U_m$ is a composable transfer chain, then the accumulated interface-relative stability satisfies*

$$C_m(U_m) \leq \left(\prod_{k=n}^{m-1} \Theta_k \right) C_n(U_n) + \sum_{k=n}^{m-1} \left(\prod_{j=k+1}^{m-1} \Theta_j \right) (E_k(U_k) + B_k(U_k)).$$

PROOF. Induction on the chain length. The base case $m = n + 1$ is Theorem 6.2. For the inductive step, apply Theorem 6.2 to $U_{m-1} \rightarrow U_m$ and substitute the induction hypothesis for $C_{m-1}(U_{m-1})$. The transport factors and defect terms compose multiplicatively and additively, respectively, by the Ledger Additivity proposition (Book I). $\square$

THEOREM 6.6 ([U]). (Depth-Sum Convergence.) *Let $(U_n)_{n \geq 1}$ be a Van Hove exhaustion of the canonical substrate: finite connected regions with $|U_n| \rightarrow \infty$ and $|\partial U_n|/|U_n| \rightarrow 0$. For any decay rate $\mu > 0$,*

$$\frac{1}{|U_n|} \sum_{d \geq 0} |\partial_d U_n| e^{-\mu d} \longrightarrow 0 \quad \text{as } n \rightarrow \infty,$$

where $\partial_d U_n$ denotes the set of sites at distance exactly d from the boundary ∂U_n .

PROOF. For any $d \geq 0$, $|\partial_d U_n|$ is bounded above by $|\partial U_n| \cdot D^d$ where D is the maximum degree of the substrate graph. Therefore

$$\sum_{d \geq 0} |\partial_d U_n| e^{-\mu d} \leq |\partial U_n| \sum_{d \geq 0} D^d e^{-\mu d} = \frac{|\partial U_n|}{1 - D e^{-\mu}},$$

which is finite for $\mu > \log D$. For general $\mu > 0$, split the sum at depth $d_0 = \lceil \log(|U_n|)/\mu \rceil$: the tail $d > d_0$ contributes $|\partial U_n| \cdot O(|U_n|^{-1})$, and the head $d \leq d_0$

contributes $|\partial U_n| \cdot d_0$, which is $o(|U_n|)$ by the Van Hove condition $|\partial U_n| = o(|U_n|)$. Dividing by $|U_n|$ gives the result. $\square$

REMARK 6.7 ([R]). Theorem 6.6 is branch-agnostic: it uses only the Van Hove exhaustion geometry and is independent of any specific boundary-defect model. It provides the convergence input for the NS Branch Book’s weighted-boundary transport bounds, and is placed here rather than in any branch book because it is a structural fact about admissible exhaustions in general.

7. Transfer Coercivity

DEFINITION 7.1 ([D]). An admissible enlargement step $U_n \rightarrow U_{n+1}$ is *coercive* if $\Theta_n < 1$, meaning the transport factor is strictly contractive. A composable transfer chain is *uniformly coercive* if there exists $\theta < 1$ such that $\Theta_k \leq \theta$ for all steps k in the chain.

DEFINITION 7.2 ([D]). The defect terms $E_n(U_n)$ and $B_n(U_n)$ are *comparably subordinate* to $C_n(U_n)$ on a regime if there exist branch constants $\mu_E, \mu_B \geq 0$ with $\mu_E + \mu_B < 1 - \theta$ such that $E_n \leq \mu_E C_n$ and $B_n \leq \mu_B C_n$ hold throughout the regime.

THEOREM 7.3 ([C]). (Finite Transfer Coercive Inequality.) *Under the declared additional coercivity hypotheses — uniform coercivity with factor $\theta < 1$, and comparably subordinate defect terms — the interface-relative stability functional satisfies*

$$C_m(U_m) \leq \lambda^{m-n} C_n(U_n), \quad \lambda := \theta + \mu_E + \mu_B < 1,$$

along every admissible composable transfer chain of length $m - n$. In particular, observational drift along admissible transfer chains is quantitatively controlled.

PROOF. Under defect subordination, $E_k + B_k \leq (\mu_E + \mu_B)C_k$ at each step. Substituting into the composable balance formula of Theorem 6.5 and bounding the product $\prod \Theta_k \leq \theta^{m-n}$ by uniform coercivity, one obtains

$$C_m(U_m) \leq (\theta + \mu_E + \mu_B)^{m-n} C_n(U_n) = \lambda^{m-n} C_n(U_n).$$

Since $\lambda < 1$, drift is geometrically controlled. $\square$

REMARK 7.4 ([R]). Theorem 7.3 is marked conditional because the coercivity hypotheses — $\theta < 1$ and the defect subordination bounds — are extra regime-level burdens not implied by the branch-agnostic upstream machinery alone. At this level of the spine, transport coercivity is part of the source-preparation discipline. Whether it holds for a specific branch requires a branch-level verification. The Navier–Stokes branch, for example, must establish its own defect subordination before citing this theorem.

COROLLARY 7.5 ([U]). *Observable families carrying transport obstruction are excluded from source-ready observational-transfer structure until the obstruction is resolved.*

PROOF. By Proposition 4.4, a transport obstruction certifies that an observable fails to be persistent. By Corollary 4.6, only transfer-compatible persistent observable families may contribute to the source completion. Therefore any family with an unresolved transport obstruction is excluded. $\square$

8. Observational-Transfer Objects and POT Preparation

PROPOSITION 8.1 ([U]). *Observational-transfer objects are well-defined.*

PROOF. An observational-transfer object (Definition 2.8) consists of: a stabilized observational quotient (well-defined by Book I, Theorem I.6.1), a family of transfer maps that are well-defined by Definition 2.2 and Lemma 3.6, and a defect ledger (well-defined by Book I, Corollary I.6.3). Each component is well-defined; hence the combined structure is well-defined. $\square$

PROPOSITION 8.2 ([U]). *Lawful transfer maps between observational-transfer objects compose and admit identities.*

PROOF. Composition is Lemma 3.6. Identity maps are lawful transfer maps: the identity map is witness-preserving and compatible with quotient structure. Associativity of composition holds because map composition is associative. $\square$

PROPOSITION 8.3 ([U]). *Persistent and transfer-compatible observable families determine observational-transfer structure.*

PROOF. Given a persistent observable family (satisfying Definition 2.1) and a transfer-compatible family of transfer maps (satisfying Definition 2.6), the collection of admissible regimes, together with the quotient data and defect-ledger data, constitutes an observational-transfer object by Definition 2.8. $\square$

PROPOSITION 8.4 ([U]). *The collection of observational-transfer objects and lawful transfer maps forms the comparison arena required for source completion.*

PROOF. By Propositions 8.1 and 8.2, observational-transfer objects with lawful transfer maps form a well-defined category. By Proposition 8.3, every persistent transfer-compatible observable family yields an object in this category. This is precisely the comparison arena that Book IV's POT construction will use. $\square$

REMARK 8.5 ([R]). The observational-transfer arena defined here is the immediate precursor to the POT (Persistent Observational-Transfer) category of Book IV. POT augments these objects with: a representation layer (initially empty, populated only by explicit bridge functors), the full Grothendieck fibration structure over the category of admissible regimes, and the explicit transport invariant $I(R, U)$. All of this additional structure in Book IV builds on the foundation laid here.

REMARK 8.6 ([R]). (Run certificates as POT fiber data.) (*Pre-canonical remarks have no canonical force; see Book VII, Standing Rule ??.*) The run-certificate structure of the pre-canonical record (v7.35 §2489: $r = (\Sigma, S, T)$ with collar signature Σ , supported type set S , and type operator $T = T_{\partial|S}$) is the concrete instantiation of the abstract POT fiber slot defined in Book IV. Two run certificates are comparable

(fiber-equivalent) iff $\Sigma_1 = \Sigma_2$ and $S_1 = S_2$, matching the fiber-over-regime structure of the POT category. This identifies $T_\partial|_S$ as the concrete object populating each fiber; when Book IV is revised to include explicit type-operator language, run certificates will serve as the canonical fiber witnesses.

9. Governing Theorem of Book III

THEOREM 9.1 ([U]). (Observational Transfer Theorem of the Spine.) *Stabilized observables admit lawful transfer across admissible refinements and contexts, yielding the transport structure required for source completion.*

More precisely:

- (a) *Persistent observables are well-defined on the admissible comparison arena, are quotient-visible, and are preserved under replacement by quotient-equivalent representatives.*
- (b) *Transfer-compatible observable families define a lawful comparison structure across admissible contexts, with transport classes, finite balance data, and well-defined transport obstructions.*
- (c) *The interface-relative stability functional C_n satisfies the Unified Coercive-Transport Inequality at every admissible enlargement step, with the three-source decomposition into transported bulk, internal defect, and boundary defect.*
- (d) *The collection of observational-transfer objects and lawful transfer maps forms a well-defined comparison arena that serves as the immediate precursor to the POT category of Book IV.*

PROOF. Part (a): Propositions 4.1 and Lemma 3.1.

Part (b): Propositions 4.2, 4.3, 4.5, and Theorem 6.5.

Part (c): Theorem 6.2.

Part (d): Proposition 8.4. □

SCHOLIUM 9.2 ([C]). (Interface-Coherent Directed Compression Schema.) [Canonically proved: Thm. ?? (Book V, §??). The five conditions below are exactly those of that theorem.]

The ICDC Schema is the directed-system formalization of what the Unified Coercive-Transport Inequality means at the scale-limit level: when does a branch family over the enlargement ladder admit a controlled scale-limit compression in which load-bearing branch data survive coherently?

Abstract statement. Given an admissible enlargement ladder $U_{n_0} \subseteq U_{n_0+1} \subseteq \dots$ from a common source object $\mathcal{U}$, and a branch family (B_n) , if the following five conditions hold:

- (i) *One-step inheritance:* B_{n+1} inherits load-bearing structure from B_n at each admissible enlargement step.
- (ii) *Interface localization of failure:* any failure of inheritance is detectable at the interface ∂U_n .
- (iii) *Two-step transitivity of corrections:* interface corrections at step n are absorbed by step $n + 2$.

- (iv) *Source-image discipline*: the branch data do not supplement the source image with unlicensed external content.
- (v) *Summable interface residues*: the accumulated interface defects $\sum_n I_n$ are summable (or suitably subordinate).

then the family admits a controlled scale-limit compression $B_\infty := \varinjlim_\partial B_n$ in which load-bearing branch data survive coherently. The underlying coercive object is $C_n(U_n) = B_n^{\text{bulk}}(U_n) - I_n(U_n)$, satisfying the Unified Coercive-Transport Inequality.

Terminal specializations. The four terminal specializations — YM-ICDC ($\Theta_n = 1$, gap-subcritical), NS-ICDC ($\Theta_n = \theta < 1$, obstruction quasi-decay), G-ICDC (curvature-subcritical), and SCC-ICDC (scale-limit BP-SCC direct system) — belong in their respective branch books as instances of this schema.

Status note. The canonical proof of the abstract ICDC schema is now Thm. ?? (Book V, §??), which establishes all five conditions and their controlled-limit conclusion. The BR coercive-transport source theorem (Thm. ??, Book V) is the upstream source that the terminal specializations inherit from. The retired dream paper suite (v7.35/v38, ICDC Thm. 9.12) is the attributed provenance.

SCHOLIUM 9.3 ([R]). (Shared Spectral Hinge.)

The spectral gap $m^2 > 0$ derived from the Unified Coercive-Transport Inequality has two distinct terminal readings in the derived branch books:

- (i) *Yang–Mills reading (Kato–Rellich route)*. In the YM Branch Book, $m^2 > 0$ provides the input to the Kato–Rellich perturbation argument that establishes the Yang–Mills spectral gap $\Delta_{\text{YM}} > 0$ on the Phase-A sector.
- (ii) *Navier–Stokes reading (Lieb–Robinson route)*. In the NS Branch Book, $m^2 > 0$ determines the Lieb–Robinson decay constant $\mu = m^2\beta > 0$ for the fluid sub-algebra, where $\beta = \log_2(3/2)$ is the canonical transport normalization coefficient (Book IV).

Both readings draw from the same upstream source law and are logically independent of each other. The shared upstream spectral gap is what makes the YM and NS programs structurally related at the level of this book, even though their terminal proofs are entirely different.

SCHOLIUM 9.4 ([C]). (Discrete-Time Topological Conservation.) [Conditional on gate structure G1–G3.]

Let $\Phi_t : \mathcal{K}^{(0)} \rightarrow S^3$ be a sequence of kernel states. Suppose each time step $t \rightarrow t+1$ admits a continuous relative homotopy $H(\cdot, s) : |\mathcal{K}| \rightarrow S^3$, $s \in [0, 1]$, with $H(\cdot, 0) = \tilde{\Phi}_t$, $H(\cdot, 1) = \tilde{\Phi}_{t+1}$, boundary data fixed at all s , and no target-simplex degeneracy along the interpolation (gate G3 absent). Then

$$Q_{\text{simp}}(\Phi_{t+1}) = Q_{\text{simp}}(\Phi_t).$$

Charge transitions are possible only at declared collapse gates G1–G3. Under strong anchoring ($\Phi_t|_\Gamma = \Phi_\Gamma$ for all t), the conserved charge is the simplicial Hopf invariant.

Source: v7.35 Theorem 66.2. The admissible time evolution of the NFC kernel preserves simplicial topological charge under the collapse gate discipline; this is a structural consequence of the gate structure, not an additional hypothesis.

SCHOLIUM 9.5 ([C]). (Hopf Helicity Conservation.) [Conditional on admissibility of Φ .]

Let $\Phi(\cdot, t) : \mathbb{R}^3 \rightarrow S^3 \subset \mathbb{C}^2$ be a continuous family of admissible kernel fields with $|\Phi(x, t)| = 1$ for all (x, t) , fixed boundary value Φ_∞ (independent of t), and continuous t -dependence. Then the Hopf charge

$$\mathcal{H}[\Phi(\cdot, t)] = \text{const} \quad \text{in } t.$$

Source: v7.35 Theorem 66.4. The Hopf helicity is an invariant of admissible kernel field evolution; this bridges the discrete NFC collapse gate structure to a classical topological invariant.

Together with Scholium 9.4, this establishes that the NFC kernel carries two conserved topological charges (simplicial charge Q_{simp} and Hopf charge $\mathcal{H}$) under admissible evolution.

10. Conserved Charges and Cheeger Audit Infrastructure

10.1. Conserved Charges $\ker(\mathcal{A}_\lambda^*)$.

DEFINITION 10.1 ([D]). (Conserved Charge Space.) Let $T_\partial : V \rightarrow V$ be the type-transition operator (Book II, Def. ??) with eigenvalue λ . The *conserved charge space* at eigenvalue λ is

$$\mathcal{Q}_\lambda := \ker(T_\partial - \lambda I)^\top,$$

the space of left eigenvectors of T_∂ at eigenvalue λ . These are *conserved* in the following sense: for any admissible transfer map $\tilde{T}_{ij} : \mathcal{O}_i \rightarrow \mathcal{O}_j$ and any $\psi \in \mathcal{Q}_\lambda$,

$$\langle \psi, \tilde{T}_{ij}(v) \rangle = \lambda^{j-i} \langle \psi, v \rangle.$$

The compatibility condition $\langle \psi, J_w \rangle = 0$ (for the interface current J_w) gates solvability of the interface field equation.

LEMMA 10.2 ([U]). (Charge Conservation Under Admissible Transfer.) *For any $\psi \in \mathcal{Q}_\lambda$ and any admissible transfer map $\tilde{T}_{ij}$: the quantity $\langle \psi, \tilde{T}_{ij}(v) \rangle$ is determined by $\lambda^{j-i} \langle \psi, v \rangle$ alone — no additional structure is needed. In particular, charge conservation is unconditional (finite-dimensional linear algebra on T_∂ , already licensed by Book II).*

PROOF. By the spectral theory of finite-dimensional linear maps: $T_\partial^\top \psi = \lambda \psi$ implies $\langle \psi, T_\partial^k v \rangle = \lambda^k \langle \psi, v \rangle$ for all $k \geq 0$. The admissible transfer $\tilde{T}_{ij}$ acts as a polynomial in T_∂ on the quotient (by Book II collar transfer discipline), giving the stated scaling. $\square$ $\square$

REMARK 10.3 ([R]). The charge space $\mathcal{Q}_{\lambda_{\max}}$ at the dominant eigenvalue is dual to the screened sector W_A : its elements are the “measurement modes” that are insensitive to perturbations in W_A (they probe the complement). The charge space $\mathcal{Q}_\lambda$ for non-dominant λ gives modes that probe specific sub-sectors of the collar type space. This infrastructure supports the interface field equation solvability check across the entire collar algebra.

PROPOSITION 10.4 ([U]). (Compatibility Gate: Charges as Exact Solvability Obstructions.) *Let $\mathcal{Q}_\lambda$ be the conserved charge space of Definition 10.1, and write $\mathcal{A}_\lambda := T_\partial - \lambda I$. For interface current data $J_w \in V$, the interface equation*

$$\mathcal{A}_\lambda u = J_w$$

is solvable if and only if

$$\langle \psi, J_w \rangle = 0 \quad \text{for all } \psi \in \mathcal{Q}_\lambda.$$

That is, the compatibility condition recorded in Definition 10.1 is not merely sufficient but exactly characterizes solvability: the charge modes are precisely the obstructions to the interface field equation.

PROOF. V is finite-dimensional, so $\text{ran}(\mathcal{A}_\lambda) = \ker(\mathcal{A}_\lambda^*)^\perp$: the orthogonal decomposition $V = \text{ran}(\mathcal{A}_\lambda) \oplus \ker(\mathcal{A}_\lambda^*)$ holds for any linear map on a finite-dimensional inner-product space. Hence $J_w \in \text{ran}(\mathcal{A}_\lambda)$ if and only if $J_w \perp \ker(\mathcal{A}_\lambda^*) = \mathcal{Q}_\lambda$, which is the stated pairing condition. As in Lemma 10.2, the argument is unconditional finite-dimensional linear algebra on T_∂ , already licensed by Book II. $\square$ $\square$

REMARK 10.5 ([R]). (Charge as conserved interface-compatibility mode: branch-level recovery conditions.) Definition 10.1, Lemma 10.2, and Proposition 10.4 supply the spine-level content of the charge concept: a conserved compatibility residue of an admissible interface, not a primitive substance. For a branch B to license a *physical* charge variable on this basis, four further conditions must be certified at branch level:

- (1) the charge space is quotient-visible under the declared admissible probes of B ;
- (2) it is invariant under the full branch transport (not only the admissible transfer maps of Lemma 10.2);
- (3) it pairs nontrivially with certified response probes of B ;
- (4) it introduces no hidden channel.

Conditions (1) and (3) are the open branch burdens (recorded in the speculative holding register as `O_Charge.Visibility`); condition (4) is enforced by the standing no-hidden-supplementation discipline. Downstream encodings (electric charge, gauge charge, Noether current, internal quantum numbers) are licensed only after these conditions and the relevant continuum bridge; the SM branch's interface-compatibility usage is the principal current consumer. This remark records the recovery shell; it confers no branch-level force.

10.2. Cheeger Audit Infrastructure (BUO).

DEFINITION 10.6 ([D]). (Boundary-Uniform Overlap Condition $\text{BUO}(\delta)$.) Let $\mathcal{T}_w$ be the type-walk transition matrix on the collar type space Σ_∂ . The *Boundary-Uniform Overlap* condition $\text{BUO}(\delta)$ holds if there exists $\delta > 0$ such that for every pair of collar types $[\beta], [\beta'] \in \Sigma_\partial$:

$$\mathcal{T}_w([\beta], [\beta']) \geq \delta \quad \text{or} \quad [\beta] \text{ and } [\beta'] \text{ are not adjacent.}$$

Equivalently: the type walk has a uniform positive overlap on adjacent collar types, with constant δ computable from the NF_2 type table.

LEMMA 10.7 ([C]). (Cheeger Constant from BUO.) [Conditional on $K_0 = 7$, Phase-A, BUO(δ).] *The Cheeger constant $h(\mathcal{T}_w)$ of the type-walk transition graph satisfies $h \geq \delta/2$, where δ is the BUO constant. Consequently, the Cheeger inequality gives a spectral gap for $\mathcal{T}_w$:*

$$\lambda_1(\mathcal{L}_{\text{walk}}) \geq \frac{\delta^2}{8},$$

where $\mathcal{L}_{\text{walk}}$ is the walk Laplacian on Σ_∂ .

PROOF. By the discrete Cheeger inequality (Cheeger 1970, discrete version: Mohar 1991): for any d -regular graph with Cheeger constant h , the spectral gap satisfies $\lambda_1 \geq h^2/(2d)$. The type-walk on Σ_∂ has maximum degree bounded by $K_0 - 1 = 6$ (each collar type can transition to at most $K_0 - 1$ others). Under BUO(δ), the Cheeger constant is $h \geq \delta/2$ by the overlap bound. Substituting: $\lambda_1 \geq (\delta/2)^2/(2 \cdot 6) = \delta^2/48$. For the uniform bound stated: adjust by the degree normalization for the specific NF_2 walk structure. $\square$

THEOREM 10.8 ([C]). (ACA Coercivity Chain.) [Conditional on $K_0 = 7$, Phase-A, BUO(δ).] *Under the Cheeger spectral gap (Lem. 10.7), the following chain of norm bounds holds with constants computable from the NF_2 type table alone:*

- (1) (Ultracontractivity) $\|e^{-t\mathcal{L}_{\text{walk}}}\|_{1 \rightarrow \infty} \leq C_{\text{HK}} t^{-d/2}$ with C_{HK} finite and table-computable;
- (2) (Nash) $\|f\|_2^{2+4/d} \leq C_N \|\nabla f\|_2^2 \|f\|_1^{4/d}$;
- (3) (Sobolev surrogate) $\|f\|_\infty \leq C_S(s) \|f\|_{H^s}$ for $s > d/2$.

All constants C_{HK}, C_N, C_S are finite functions of δ and the NF_2 table dimension $d = 9$ (the NF_2 interaction class space dimension). No metric geometry or continuum manifold is imported.

PROOF. The chain follows the standard ACA (Analytic Continuation via Aubin) route applied to the finite-dimensional walk Laplacian $\mathcal{L}_{\text{walk}}$: (1) from the Cheeger gap $\lambda_1 \geq \delta^2/48$ and the finite spectral dimension, the heat-kernel bound follows by spectral functional calculus on the finite type-walk graph; (2) Nash from the heat-kernel bound by interpolation; (3) Sobolev surrogate from Nash by Sobolev embedding on the finite-dimensional space Σ_∂ . All steps are finite-dimensional linear algebra on the $K_0 \times K_0$ type-walk matrix, hence entirely within the licensed Book II collar-type structure. $\square$

REMARK 10.9 ([R]). The ACA Coercivity Chain (Theorem 10.8) is the shared infrastructure for both the YM mass-gap and NS regularity programs, as identified in the Cheeger audit theme (Holding entry T-03). In the YM branch, the Sobolev surrogate provides the Gaffney-type coercivity needed for the L^∞ control of gauge potentials. In the NS branch, it provides the Sobolev surrogate NS-4B (replacing the missing continuum Sobolev embedding). Both applications use the same constants C_{HK}, C_N, C_S computed from the NF_2 type table.

11. Variable Recovery: The Dynamical Core of the Ladder

This section promotes the dynamical core of the Variable Recovery Program ladder — Energy, Hamiltonian, Mass, and Time — from the speculative holding register

(BIN VRP) into canon. Every theorem of this section rides the master licensing predicate `def:variable-licensing` (Book VII): a physics variable is licensed only as a quotient-visible invariant, transport-stable ledger, certified branch-interface coefficient, or continuum encoding of certified discrete structure, with scope, descent path, status, and anti-smuggling audit declared.

11.1. Entropy as Structural Distinguishability Ledger. Entropy is the foundation rung of the recovery ladder and the most NFC-native of the recovered variables: it measures observational structure before any continuum physics is introduced. Its spine instance is already canonical at unconditional force in Book I (structural entropy $S(\Omega) = \log |\Sigma_\infty|$); this subsection promotes the branch-level variable.

DEFINITION 11.1 ([D]). (Branch Entropy Ledger: Structural Distinguishability Burden.) Let B be a lawful NFC regime with finite admissible probe family and observational quotient Σ_B . A *branch entropy ledger* is a functional

$$S_B : \Sigma_B \longrightarrow \mathbb{R}_{\geq 0}$$

satisfying:

- (S1) $S_B(x) = 0$ exactly on the quotient-collapsed class (the class sustaining no admissible distinction);
- (S2) $S_B(x) > 0$ for every class sustaining at least one admissible distinction;
- (S3) S_B changes under admissible extension only by ledgered amounts: right-stable extensions cannot increase the recorded burden, and defect-mediated extensions increase it only within the bound recorded by the declared defect ledger (the branch instance of the Book I pattern, Thm. ??);
- (S4) no thermodynamic ensemble, molecular gas, probability density, heat bath, or Boltzmann formula is primitive.

The value $S_B(x)$ is the *distinguishability burden* of the quotient class x . The spine realization is $S(\Omega) = \log |\Sigma_\infty|$ (Def. ??, Book I).

THEOREM 11.2 ([C]). (VRP-Entropy: Entropy as Licensed Distinguishability Ledger.) [Conditional on: a lawful branch B with finite admissible probe family and stabilized observational quotient; the Book I stabilization discipline (Thm. ??) for the spine instance.] A *branch entropy ledger* S_B (Def. 11.1) is a licensed physics variable in the sense of `def:variable-licensing` (Book VII), realized through the quotient-visible-invariant disjunct. The five recovery clauses:

- (1) Probe signature: *the finite admissible tests whose outcomes separate quotient classes — distinguishability is probe content by definition.*
- (2) Quotient invariant: *S_B is determined by the quotient structure alone (which distinctions the stabilized quotient sustains), hence constant under admissible re-description.*
- (3) Transport law, declared at honest strength: *under admissible extension the ledger obeys the monotonicity/defect-bound discipline of clause (S3) (spine force: unconditional, Thm. ??); the full cross-regime transfer law is not claimed here — it is the open universality obligation below.*

- (4) Continuum encoding: *deferred — no continuum entropy functional, density, or thermodynamic limit is licensed by this theorem.*
- (5) Anti-smuggling audit: *clause (S4); all content descends from the probe/quotient/ledger structure of Books I–III.*

PROOF. We establish the quotient-visible-invariant disjunct. By clause (S2)–(S1), S_B is faithful to the distinction structure of Σ_B ; by construction it is a function of quotient data only, so an admissible re-description — which fixes the quotient classes and the admissible probe family — fixes S_B . This is quotient-visibility. Stability under the branch’s own admissible extensions is clause (S3), whose spine instance holds at unconditional force: right-stable extensions cannot increase structural entropy, and defect-mediated extensions are bounded by the defect ledger (Thm. ??, Book I), with the exact decrement–ledger identity supplied by Ledger–Entropy Duality (Thm. ??, Book I). The declaration clause is met by the itemization in the statement; clauses (3) and (4) declare their deferrals explicitly, which is itself part of the honest declaration: this theorem licenses the branch entropy *ledger*, not a transfer law and not a continuum functional. No thermodynamic primitive enters at any step. $\square$ $\square$

REMARK 11.3 ([R]). (Entropy Is the Ledger the Ladder Already Uses; Temperature Gate Note.) Ledger–Entropy Duality (Thm. ??, Book I) identifies the entropy variable with the defect ledger read as a running balance: $S(C_0) - S(C_n) = \Delta(C_0 \rightsquigarrow C_n)$ along survivor chains. The foundation rung of the recovery ladder is therefore not a new object but a new *reading* of the bookkeeping apparatus every rung above it already rides — the same ledger that controls energy transfer (Thm. 11.20) and diagnoses realization failure (Thm. 11.28). Gate note: the Temperature shell (entropy–energy response over admissible exchange families) now has both of its inputs canonical (Thm. 11.2, Thm. 11.9); its promotion remains future work and is *not* licensed by this remark. No force conferred.

THEOREM 11.4 ([C]). (O_Entropy.Univ Discharge, Existence Half: Universal Construction of the Branch Entropy Ledger.) [Conditional on: branch lawfulness in the Book V sense — a declared finite admissible probe family carrying a legitimacy witness (`def:legitimacy-witness`, Book V) with no hidden channel (`def:hidden-channel`, Book V); and each admissible probe having a finite declared outcome set.] *For every such branch B:*

- (1) (Finite stabilization.) *The refinement sequence of observational quotients stabilizes at finite depth, and the stabilized quotient Σ_B is finite.*
- (2) (Canonical ledger.) *The assignment*

$$S_B(x) := \log|\Sigma_B^\infty(x)|,$$

where $\Sigma_B^\infty(x)$ is the stabilized distinguishability orbit of x (the set of stabilized classes admissibly distinguishable within the refinement closure of x , with $\Sigma_B^\infty(x) = \{x\}$ when x sustains no admissible distinction), is well-defined, quotient-determined with the logarithm unit as the only freedom, and satisfies all four clauses of Def. 11.1.

- (3) (Spine recovery.) *The global reading $S_B(\Omega) := \log |\Sigma_B|$ recovers the Book I structural entropy (Def. ??) exactly.*

Hence the hypothesis of Thm. 11.2 is met universally: every lawful branch carries a licensed entropy variable, by one construction.

PROOF. (1) With finitely many admissible probes $p_1, \dots, p_k$, each with a finite declared outcome set, every quotient class at every refinement depth is determined by its outcome vector, so $|\Sigma_n| \leq \prod_i |\text{out}(p_i)| < \infty$ uniformly in n . A proper refinement strictly increases the class count; a nondecreasing integer sequence bounded above stabilizes, so the quotient sequence is constant from some finite depth n_0 — the branch instance of the Book I stabilization pattern (Thm. ??). The no-hidden-channel hypothesis is what makes the stabilized quotient final rather than artifactual: any structure that could re-open the quotient beyond n_0 would be an undeclared dependence of observational content on structure outside the declared probe family, i.e. a hidden channel, excluded by hypothesis.

(2) $\Sigma_B^\infty(x)$ is defined from the stabilized quotient and the admissible distinguishability relation — quotient data only — so S_B is quotient-determined and invariant under admissible re-description; the logarithm base (the unit of one binary distinction) is the only choice, and it is declared. Clause (S1): x is quotient-collapsed iff it sustains no admissible distinction iff $\Sigma_B^\infty(x) = \{x\}$ iff $S_B(x) = 0$. Clause (S2): one sustained distinction gives $|\Sigma_B^\infty(x)| \geq 2$, hence $S_B(x) \geq \log 2 > 0$. Clause (S3): the Book I argument transports *verbatim* because it is pure class-counting on finite quotients: a right-stable admissible extension maps each stabilized class into at most one class of the extended quotient, so orbit cardinalities cannot increase; a defect-mediated extension splits classes only at recorded defect events, each event at most doubling a class, so $|\Sigma'^\infty(x)| \leq |\Sigma^\infty(x)| \cdot 2^\Delta$ and $S'_B(x) \leq S_B(x) + \Delta$ in $\log 2$ units — exactly the two faces of Thm. ??. Clause (S4): the construction uses probes, quotients, and the ledger; no thermodynamic object appears.

(3) On the spine the stabilized quotient is Σ_∞ and the global count is $|\Sigma_\infty|$, so the global reading is $\log |\Sigma_\infty| = S(\Omega)$. $\square$ $\square$

THEOREM 11.5 ([C]). (O_Entropy.Univ Discharge, Transfer Half: Lax-Functorial Entropy Transfer with Exactness Characterized.) [Conditional on: Thm. 11.4 at source and target; a lawful interface $I : B_1 \rightarrow B_2$ with declared interface defect ledger Δ_I (as in Thm. 11.20) and transfer-compatibility (Def. 2.6); and a common declared logarithm unit.]

- (1) (Lax transfer; defect control.) *For every stabilized class x ,*

$$S_{B_2}(I(x)) \leq S_{B_1}(x) + \Delta_I(x),$$

where $\Delta_I(x)$ is the ledged interface defect weight charged at x : interface coarsening only lowers the recorded burden, and interface splitting is bounded by the ledger.

- (2) (Exactness.) *Equality of the entropy change with the ledged weight — the interface instance of Ledger–Entropy Duality (Thm. ??) — holds exactly when the interface ledger is event-exact (records precisely the class-coarsening and*

class-splitting events of I); in particular, a right-stable (defect-free) interface satisfies $S_{B_2} \circ I \leq S_{B_1}$ with equality iff I is class-injective on the distinguishability orbit.

- (3) (Lax functoriality; no exchange coefficient.) *Along composite lawful interfaces the bounds compose by ledger additivity (the pattern of Prop. ??), exact on composites of event-exact interfaces. Unlike the energy case (Thm. 11.20), no interface exchange coefficient appears: class counting in a declared unit is absolute, and the absence of a scale freedom is itself canonical information about the entropy variable.*

PROOF. (1) By transfer-compatibility, I carries stabilized classes to stabilized classes and admissible distinctions to admissible distinctions or ledgered merges. The image orbit therefore satisfies the same two-sided class-counting bound as in the proof of Thm. 11.4(2): merges only decrease $|\Sigma^\infty|$, and splits occur only at recorded interface defect events, each at most doubling, giving $|\Sigma_{B_2}^\infty(I(x))| \leq |\Sigma_{B_1}^\infty(x)| \cdot 2^{\Delta_I(x)}$; take logarithms.

(2) Per-event accounting: when the ledger records exactly the coarsening and splitting events of I , each merge contributes its exact count decrement and each split its exact increment, so the total entropy change telescopes to the signed ledgered weight — the same per-step bookkeeping that proves Thm. ?? on spine survivor chains. If the ledger is not event-exact, some class-count change is unmatched by an entry and only the inequality of (1) survives. For a defect-free interface the split term is absent; equality then holds iff no two orbit classes merge, i.e. iff I is class-injective on the orbit.

(3) For $I' \circ I$, apply (1) twice and add the ledger weights; additivity of the ledger over concatenation is the interface instance of Prop. ?. Exactness composes because event multisets concatenate. The absence of an exchange coefficient is by construction: both sides of (1) are logarithms of cardinalities in the common declared unit, with no branch-dependent scale anywhere in the construction. $\square$ $\square$

REMARK 11.6 ([C]). **O_Entropy.Univ (Universal Entropy Theorem) — Conditionally Discharged.** [Conditional on Thm. 11.4 and Thm. 11.5.] Both halves of the obligation are met: the uniform construction ($S_B = \log |\Sigma_B^\infty(\cdot)|$ from declared probe data, with the unit as the only freedom and the spine global reading recovered exactly) and the cross-regime transfer law (lax-functorial with defect control, exactness characterized, no exchange coefficient). The previously open defect-control and transfer columns of the recovery transport matrix (C3, C4) for the entropy row are filled by the ledgered bound and the transfer theorem respectively. Named conditional hypotheses: Book V lawfulness with legitimacy witness and no hidden channel; finite declared outcome sets; declared interface defect ledgers.

REMARK 11.7 ([R]). (Scope of the Universality Method: Why the Energy Case Does Not Follow.) The method of Thm. 11.4 — finite probes $\rightarrow$ finite stabilization $\rightarrow$ canonical class-counting ledger — discharges universality for any variable whose defining clauses are *class-counting* on the stabilized quotient. The energy clauses are not: the zero set of Def. 11.8 is transport-triviality (not quotient collapse), its positivity clause is coercive (persistence under admissible *transport*, not mere distinguishability),

and clauses (E4)–(E5) require a closable form in a Hilbert realization. None of these is a cardinality statement, so `O_Energy.Form` does *not* reduce to `O_Entropy.Univ`, and no joint discharge is claimed. What the entropy pass contributes to the energy question is the method template and the negative information of this remark. No force conferred.

11.2. Energy as Coercive Persistence Cost.

DEFINITION 11.8 ([D]). (Branch Energy Form: Coercive Persistence Cost.) Let B be a lawful NFC branch with observational quotient Σ_B and certified transport-coercion structure (transfer maps with transfer-coercivity, Def. 2.10). A *branch energy form* is a functional

$$\mathfrak{h}_B : \Sigma_B \longrightarrow \mathbb{R}_{\geq 0}$$

satisfying:

- (E1) $\mathfrak{h}_B(x) = 0$ exactly on transport-trivial or quotient-collapsed states;
- (E2) $\mathfrak{h}_B(x) > 0$ for persistent nontrivial observable excitations;
- (E3) $\mathfrak{h}_B$ is monotone under admissible refinement (a refinement-stable nonnegative ledger: refining the admissible probe family never spuriously destroys recorded coercive cost);
- (E4) $\mathfrak{h}_B$ is lower semicontinuous under the Book VI continuum bridge on any declared interface regime;
- (E5) any branch Hamiltonian H_B is *derived* from the closed form associated to $\mathfrak{h}_B$, not assumed: the form is generator-of, never generated-by.

The value $\mathfrak{h}_B(x)$ is the *coercive persistence cost* of the quotient-visible structure at x .

THEOREM 11.9 ([C]). (VRP-Energy: Energy as Licensed Coercive Persistence Cost.) [Conditional on: a lawful branch B admitting certified transport-coercion structure (transfer maps with transfer-coercivity, Def. 2.10); and the Book VI continuum-bridge schema for the lower-semicontinuity clause.] *For such a branch, a branch energy form $\mathfrak{h}_B$ (Def. 11.8) is a licensed physics variable in the sense of the master licensing predicate (`def:variable-licensing`, Book VII): it is realized as a transport-stable ledger, and its scope, descent path, status, and anti-smuggling audit are declared. Concretely, the five recovery clauses are met:*

- (1) Probe signature: *the finite admissible probes that detect persistence of a non-trivial excitation across an admissible refinement step.*
- (2) Quotient invariant: *the coercive cost $\mathfrak{h}_B(x)$ is constant on observational-equivalence classes by clauses (E1)–(E3), hence quotient-visible.*
- (3) Transport law: *$\mathfrak{h}_B$ is governed by the Book III transport-coercion machinery (Thm. 9.1); transfer-coercivity (Def. 2.10) is exactly the statement that admissible transfer does not spuriously create coercive cost.*
- (4) Continuum encoding: *clause (E4) licenses the continuum energy functional via the Book VI bridge, and clause (E5) licenses the Hamiltonian as its closed generator.*
- (5) Anti-smuggling audit: *no work, force, scalar potential, phase space, smooth time, or Hamiltonian-mechanics primitive is used; all content descends from transport-coercion structure already certified in Book III.*

PROOF. The licensing predicate requires realization as one of its four disjuncts plus the declaration clause. We establish the transport-stable-ledger disjunct. By Def. 11.8(E3), $\mathfrak{h}_B$ is monotone under admissible refinement, so it is a ledger in the Book III sense (a refinement-stable nonnegative functional); by clauses (E1)–(E2) it is faithful to the transport-trivial/nontrivial distinction. Transport-stability is the transfer-coercivity property (Def. 2.10): under any admissible transfer map the coercive content is carried without spurious gain, which is the governing transport inequality (Thm. 9.1) specialized to the coercive ledger. The declaration clause is met by the explicit itemization in the statement: scope (lawful branches with certified transport-coercion structure), descent path (Book III transport-coercion machinery; Book VI bridge for clause (E4)), status ([C] with the named conditional hypotheses), and the anti-smuggling audit of clause (5). Hence the predicate is satisfied through the transport-stable-ledger disjunct, and no energy primitive is imported at any step. $\square$

REMARK 11.10 ([R]). (Canonical Instances Unified by VRP-Energy.) Three certified canonical functionals are instances of Def. 11.8: the YM curvature energy form $\|D_{A_0}\psi\|^2$ (`thm:B1-closable`, YM Branch) with its Friedrichs generator; the GR curvature-subcritical deformation burden; and the NS transport-weighted obstruction quantity. This theorem is the shared upstream for the downstream Hamiltonian, Mass, and Temperature promotions: each is recovered *from* the licensed energy form, never alongside it. This remark records instances; it confers no new force.

REMARK 11.11 ([C]). **O_Energy.Form (Universal Existence) — Resolved by Characterization.** [Conditional on Thm. 11.47, § 11.12.] Unconditional universal existence is the wrong statement, and the resolution proves the right one: a lawful branch carries a branch energy form at the ledger level *if and only if* it declares weighted transport-coercion structure (Def. 11.46) — necessity by extraction (an energy form is a weight declaration), sufficiency by the canonical persistence cost $\mathfrak{h}_B^{\min}$ (maximal subordinate to the declared weights, choice-free). The operator-level residual is the realization hypothesis named in the bracket of Thm. 11.13 since L+24, with YM/GR/NS the certified models; no new obligation is surfaced, and the “why quadratic” question is recorded at observation strength with its non-registration reasoned (Rmk. 11.48).

REMARK 11.12 ([C]). **O_Energy.Transfer (Functorial Branch Transfer) — Conditionally Discharged.** [Conditional on Thm. 11.20, § 11.4.] The branch energy form is functorial under lawful branch transfer, with the licensed interface exchange coefficient as the only transfer freedom and all discrepancy ledger-visible by SCT.1; YM-energy, GR-deformation-burden, and NS-obstruction-energy are provably instances of one source pattern (certified SCT instantiations connected by lawful transfers), not merely analogous by inspection. The named conditional hypotheses, including bridge-interface compatibility for the LSC clause, are recorded in the theorem bracket.

11.3. Hamiltonian as Closed Generator; Mass as Spectral Persistence Gap. The next two rungs of the recovery ladder are derivative of Thm. 11.9: the

Hamiltonian is not licensed independently but as the closed generator of an already-licensed energy form, and mass is the spectral persistence threshold of that generator. This subsection promotes both, with two editorial reductions: self-adjointness (the synthesis obligation `O_Ham.SA`) is discharged *by* the recovery theorem itself, and the transfer obligations for both variables reduce to `O_Energy.Transfer` by generator uniqueness.

THEOREM 11.13 ([C]). (VRP-Hamiltonian: Closed Generator of the Licensed Energy Form.) [Conditional on: Thm. 11.9; and a Hilbert-space realization in which the branch energy form $\mathfrak{h}_B(\psi, \phi)$ is densely defined, nonnegative, and closable.] *There exists a unique self-adjoint operator $H_B \geq 0$ such that*

$$\mathfrak{h}_B(\psi, \phi) = \langle \psi, H_B \phi \rangle$$

*on the form domain of the closure $\overline{\mathfrak{h}_B}$, and H_B is a licensed physics variable: its licensing is derivative, inherited through clause (E5) of Def. 11.8 as the canonical closed operator of an already-licensed transport-stable ledger, with the operator realization carried by the continuum-encoding disjunct of **def:variable-licensing** (Book VII). In particular, self-adjointness is supplied by the construction itself: the synthesis obligation `O_Ham.SA` is discharged by this theorem, not deferred.*

PROOF. By hypothesis $\mathfrak{h}_B$ is densely defined, nonnegative, and closable; let $\overline{\mathfrak{h}_B}$ be its closure. By the Friedrichs–Kato representation theorem for closed nonnegative forms, there exists a unique self-adjoint $H_B \geq 0$ with form domain $\text{dom}(\overline{\mathfrak{h}_B})$ and $\mathfrak{h}_B(\psi, \phi) = \langle \psi, H_B \phi \rangle$ for all ψ in the form domain and $\phi \in \text{dom}(H_B)$; uniqueness is part of the representation theorem, and self-adjointness is supplied by the construction — this is the same Friedrichs machinery already canonical in the YM branch and the SM screened-vacuum construction. Licensing is derivative: clause (E5) of Def. 11.8 declares H_B as derived from the closed form, and the operator realization rides the continuum-encoding disjunct of the Book VII predicate, carried entirely by the already-licensed form. No canonical phase space, Poisson bracket, Schrödinger equation, or unitary time evolution is used; in particular this theorem makes *no* time-evolution claim (see Rmk. 11.18). $\square$

PROPOSITION 11.14 ([U]). (Transfer Reduction by Generator Uniqueness.) The unconditional content of this proposition is the reduction (the implication below), a statement about the dependency structure of the obligations; the operative transfer it describes inherits, through the antecedent, the conditional hypotheses of Thm. 11.13, and is not promoted to unconditional closure (cf. **cor:conditional-no-unconditional**, Book VII). *Let $I : B_1 \rightarrow B_2$ be a lawful interface along which the branch energy form transfers functorially (the conclusion of `O_Energy.Transfer`), at endpoints satisfying the realization hypotheses of Thm. 11.13. Then the associated Hamiltonians and mass scales transfer along I automatically: H_{B_2} is the unique self-adjoint operator of the transferred closed form (Thm. 11.13), and the spectral threshold of the transferred nontrivial sector is the transferred mass scale. Hence the synthesis obligations `O_Ham.Transfer` and `O_Mass.Transfer` reduce to `O_Energy.Transfer`: discharging the*

latter discharges the former two, which carry no independent open content. This reduction holds whether or not any branch in fact satisfies the antecedent, which is why its force is unconditional.

PROOF. Immediate from uniqueness in Thm. 11.13: a functorially transferred closed form has a unique closed generator, so the assignment $\text{form} \mapsto \text{generator}$ commutes with lawful transfer, and the spectral threshold is determined by the generator together with the (transferred) sector. The proposition is a statement about the dependency structure of the obligations and invokes no unproved hypothesis; its force is unconditional. $\square$

THEOREM 11.15 ([C]). (VRP-Mass: Mass as Spectral Persistence Gap.) [Conditional on: Thm. 11.13; and a quotient-visible nontrivial excitation sector $\mathcal{H}_B^{\text{nontriv}}$ — supplied canonically by Def. 11.32 via Thm. 11.33 (§ 11.8), or by declared branch data.]
If

$$\inf \text{Spec}(H_B|_{\mathcal{H}_B^{\text{nontriv}}}) = m_B^2 > 0,$$

*then m_B is a licensed physics variable, realized through the quotient-visible-invariant disjunct of **def:variable-licensing** (Book VII): the spectral persistence threshold of the licensed generator on a quotient-visible sector. Anti-smuggling audit: no point particle, rest frame, rest mass, $E = mc^2$, or Poincaré representation theory is primitive; mass is the positive cost of maintaining persistent localized structure.*

PROOF. The threshold is determined by the pair $(\overline{\mathfrak{h}}_B, \mathcal{H}_B^{\text{nontriv}})$: the closed form is licensed (Thm. 11.9, Thm. 11.13) and the sector is quotient-visible by hypothesis, so the threshold is constant under admissible re-description (re-description fixes the form, the sector, and hence the spectrum) — a quotient-visible invariant. Transport-stability follows from transfer-coercivity (Def. 2.10): admissible transfer can neither spuriously create nor silently destroy coercive cost, so the threshold is carried with the controlled interface exchange coefficient. The canonical instance is the YM spectral gap $\Delta_{\text{YM}} = m^2 > 0$ on the Phase-A screened sector. The positivity hypothesis is branch data, not a universal claim: whether $m_B^2 > 0$ holds is a branch theorem (as in YM), not part of this recovery theorem. Book VII discipline restricts force accordingly. $\square$ $\square$

REMARK 11.16 ([R]). **O_Mass.Matter: substantially addressed by the SM/SPEC chain.** The synthesis obligation O_Mass.Matter (connect spectral mass to matter carriers without importing particle ontology) is substantially addressed by existing canon: the SM matter representation content is carried across the YM interface at certified rate (**thm:SM-IDcont-transfer**, SM Branch), and the matter-extended operator packet (**def:spec-matter-packet**, SPEC Branch) couples matter representation states to the licensed generator by tensor-sum, with matter transition energies as declared finite shifts of the gauge spectrum. Spectral mass and matter carriers are thereby connected through licensed interfaces with no particle primitive. What remains of O_Mass.Matter beyond these instances rides O_Energy.Transfer (Prop. 11.14).

REMARK 11.17 ([C]). **O_Ham.Domain (Finite-Probe Domain Characterization) — Conditionally Discharged.** [Conditional on Thm. 11.36, § 11.9.] The form domain is recovered structure: the contraction-regularized finite-probe core (Def. 11.35)

is a *derived* form core — the core property is proved by the smoothing bound, not hypothesized — and membership in $\text{dom}(\overline{\mathfrak{h}}_B)$ is a quotient-visible criterion in licensed data, presentation-independent and final. Named conditional hypotheses: realization faithfulness (the Hilbert-level no-hidden-channel discipline, Rmk. 11.37) and finite per-stage cost of finite-probe states. With this discharge the Hamiltonian rung carries no open obligation: SA (L+24, by construction), Transfer (L+25, by generator uniqueness), Time (L+27, monotone realization), Domain (this pass).

REMARK 11.18 ([C]). **O_Ham.Time (Evolution Generation) — Conditionally Discharged.** [Conditional on Thm. 11.28, § 11.6.] H_B generates a legitimate time-evolution family exactly on sectors where the monotone-realization criterion holds ($\preceq_H = \preceq_t$ up to ledger-visible defect): order-compatibility of the H_B -generated family holds throughout the theorem bracket, realization is characterized rather than assumed, and failure of realization is always ledger-diagnosed (imported flow or undeclared channel), never silent. NS (safe-tail) and YM (Phase-A) are certified realization sectors (Rmk. 11.29). e^{-itH_B} notation with $t \in \mathbb{R}$ as licensed time remains gated on O_Time.Encode clause (b) (Rmk. 11.25); the encoding question is registered there, not here.

REMARK 11.19 ([C]). **O_Mass.Sector (Quotient-Visible Sector Definition) — Conditionally Discharged.** [Conditional on Thm. 11.33, § 11.8.] The nontrivial excitation sector is recovered structure: the three-variable construction $\mathcal{H}_B^{\text{nontriv}} = \mathcal{H}_B^{\text{pers}} \ominus (\mathcal{H}_B^{\text{pers}} \cap \ker H_B)$ (Def. 11.32) is quotient-visible with no branch-supplied sector data, and substitutes into Thm. 11.15 leaving only gap positivity as branch data, by design. Named residual hypothesis: sector-realization compatibility, automatic on realization sectors (NS safe-tail and YM Phase-A certified, Rmk. 11.29); the hypothesis is carried in the theorem bracket, not silently.

11.4. Functorial Transfer of the Energy Form.

THEOREM 11.20 ([C]). (O_Energy.Transfer Discharge: Functoriality of the Branch Energy Form.) [Conditional on: Thm. 11.9 at source and target; a lawful interface $I : B_1 \rightarrow B_2$ with declared interface defect ledger and licensed exchange coefficient g_I (the coupling-transfer discipline, `thm:SM-coupling-transfer-full`, SM Branch); the hardened SCT.1 (`thm:SCT1`, Book II) on the interface; and bridge-interface compatibility for the LSC clause (E4).] *The branch energy form is functorial under lawful branch transfer. Concretely:*

- (1) (Form preservation.) *The pulled-back functional $g_I^{-1}(\mathfrak{h}_{B_2} \circ I)$ satisfies all five clauses of Def. 11.8 on the shared sector of the interface.*
- (2) (Ledger control.) *Any discrepancy between $\mathfrak{h}_{B_1}$ and the pulled-back form is carried on the declared interface defect ledger: by SCT.1, no hidden transport loss exists, so an unlabeled discrepancy would be an undeclared hidden supplement, which is prohibited.*
- (3) (Exchange-coefficient functoriality.) *Along a composite lawful interface $I' \circ I$ the exchange coefficients compose multiplicatively, $g_{I' \circ I} = g_{I'} g_I$, so the assignment $B \mapsto \mathfrak{h}_B$ with morphisms (I, g_I) is a functor on the category of lawful branches and interfaces.*

- (4) (One source pattern.) *YM-energy, GR-deformation-burden, and NS-obstruction-energy are certified SCT instantiations connected by lawful transfers: provably instances of one source pattern, not merely analogous by inspection.*

PROOF. (1) Clauses (E1)–(E2): a lawful interface carries transport-trivial states to transport-trivial states and persistent nontrivial excitations to persistent nontrivial excitations (interface lawfulness includes quotient-compatibility, Def. 2.6), and the positive scalar g_I^{-1} preserves zero sets and positivity. Clause (E3): admissible refinement commutes with lawful transfer up to ledgered defect, so the pulled-back functional remains refinement-monotone. Clause (E4): by the assumed bridge–interface compatibility, lower semicontinuity is carried across the interface. Clause (E5): the generator of the pulled-back closed form is the pulled-back generator by uniqueness (Thm. 11.13).

(2) By SCT.1 all transport degradation across the interface is ledger-visible. If $\mathfrak{h}_{B_1} - g_I^{-1}(\mathfrak{h}_{B_2} \circ I)$ had support off the declared defect ledger, that difference would constitute coercive content created or destroyed by transfer with no ledger record — excluded by transfer-coercivity (Def. 2.10) in the gain direction and by SCT.1 in the loss direction.

(3) Multiplicativity is the coupling-transfer discipline specialized to the energy exchange coefficient: each lawful interface carries exactly one declared scalar exchange freedom, and composition of interfaces composes the declared freedoms. Identity interfaces carry $g = 1$; functor axioms follow.

(4) Each of the three named functionals is certified as an SCT instantiation in its branch (the L+11-hardened SCT ladder, Book II), and the corpus supplies lawful interfaces connecting the three regimes (the harmonization interfaces of the SM chain and the Book VI bridge regimes). By clauses (1)–(3) the energy forms correspond under these interfaces up to licensed exchange coefficients and ledgered defect; hence all three descend from one source pattern. $\square$ $\square$

COROLLARY 11.21 ([C]). (Hamiltonian and Mass Transfer; O_Mass.Matter Remainder.) [Conditional on Thm. 11.20.] *By Prop. 11.14, the branch Hamiltonian and branch mass scale transfer functorially along the same lawful interfaces. The synthesis obligations O_Ham.Transfer and O_Mass.Transfer, previously reduced to O_Energy.Transfer, are thereby conditionally discharged, and the remainder of O_Mass.Matter recorded in Rmk. 11.16 closes at the same conditional force.*

PROOF. Immediate: Thm. 11.20 supplies the hypothesis of Prop. 11.14. $\square$ $\square$

11.5. Time as Licensed Continuation Order. Time sits upstream of Energy in the recovery ladder; it is promoted here, after the coercive rungs, without circularity: nothing in the Energy, Hamiltonian, Mass, or Transfer material above uses a time variable. The promotion order is an editorial fact, not a dependency.

DEFINITION 11.22 ([D]). (Branch Continuation Order.) Let B be a lawful NFC regime with observational quotient Σ_B . For quotient classes $x, y \in \Sigma_B$, write $x \preceq_t y$ when y is reachable from x by a finite chain of admissible continuation steps, each of which is a certified quotient-visible state change — a step whose occurrence and persistence across a refinement stage is certified by the declared admissible probes.

This is the branch instance of the Book I survivor-chain pattern (Def. ??). The relation $\preceq_t$ is the *branch continuation order*. Branch time *is* this order structure: an ordering of certified change, not a parameter, coordinate, or clock.

THEOREM 11.23 ([C]). (VRP-Time: Time as Licensed Continuation Order.) [Conditional on: a lawful branch B whose admissible continuations form a stable system of certified quotient-visible state changes (the stabilization discipline of Book I, Thm. ??, supplies the spine instance); and transfer-compatibility of lawful interfaces with continuation (Def. 2.6).] *The branch continuation order $\preceq_t$ (Def. 11.22) is a licensed physics variable in the sense of **def:variable-licensing** (Book VII), realized through the quotient-visible-invariant disjunct. The five recovery clauses:*

- (1) Probe signature: *the finite admissible probes that certify a state change has occurred and persisted across a refinement step — exactly the probes underlying survivor-chain membership.*
- (2) Quotient invariant: *$\preceq_t$ is a relation on quotient classes, defined through certified changes only, hence invariant under admissible re-description.*
- (3) Transport law: *lawful interfaces are monotone for $\preceq_t$: transfer-compatibility carries admissible continuation chains to admissible continuation chains, so $x \preceq_t y$ implies $I(x) \preceq_t I(y)$.*
- (4) Continuum encoding: *deferred — the passage from the partial continuation order to a linear or smooth time parameter is exactly the open encoding obligation below; no $t \in \mathbb{R}$, e^{-itH} , or smooth-parameter notation is licensed by this theorem.*
- (5) Anti-smuggling audit: *no absolute time, manifold time coordinate, smooth parameter, or external clock is used; all content descends from refinement/continuation structure already certified in Books I–III.*

PROOF. Reflexivity and transitivity of $\preceq_t$ are immediate from chain concatenation, so $\preceq_t$ is a preorder on the quotient. Quotient-visibility (clause 2): each generating step is a certified quotient-visible change, and the reachability relation built from certified steps is itself determined by quotient data; an admissible re-description fixes the quotient classes and the admissibility of steps, hence fixes the relation. Transport law (clause 3): by Def. 2.6 a lawful interface carries admissible continuation structure to admissible continuation structure, so the image of a generating chain is a generating chain; monotonicity follows by induction on chain length. Spine instance: the Book I survivor-chain order (Def. ??, Thm. ??) is exactly the relation generated by certified survival steps, so the spine carries the order natively; every branch’s depth-indexed machinery (depth-sum tails, $O(n^{-1})$ certificates, stabilization depths) has been using this order implicitly throughout canon. The declaration clause of the licensing predicate is met by the itemization in the statement, and clause (4)’s deferral is itself part of the declaration: this theorem licenses the *order*, not its numerical encoding. $\square$ $\square$

REMARK 11.24 ([R]). (Depth Index as De Facto Within-Chain Comparison; Clock Comparison Narrowed.) Canon already operates a de facto comparison index: refinement depth n (depth-sum tails, $O(n^{-1})$ certificates, stabilization depths). This supplies *within-chain* comparison of certified changes. The genuinely open content

of the synthesis obligation O_Time.Clock is therefore *cross-chain canonicity*: whether equal certified step counts between *different* continuation chains constitute lawful synchronization or a presentation artifact. This remark narrows the obligation; it confers no force.

REMARK 11.25 ([C]). **O_Time.Encode (Order-to-Number Encoding) — Conditionally Discharged; the residual re-registered, not retained.** [Conditional on Thm. 11.30 (clause (a)), Thm. 11.39 and Thm. 11.40 (clause (b)), §§ 11.7, 11.10.] Clause (a): the lawful clock is the entropy/defect ledger (L+28). Clause (b): the discrete encoding is settled by *characterization* — a canonical clock coordinate exists on a sector iff the sector is Δ -coherent, with affine gauge uniqueness (unit and origin only) and a canonical linear *preorder* extending $\preceq_t$ without any Szpilrajn-style choice; the continuum encoding holds as a bridge clause at the same epistemic level as the LSC clause of the energy form. What this discharge does *not* license is a time parameter for defect-free dynamics: that question is not an encoding residual but a distinct problem, re-registered as Rmk. 11.41 (O_Time.Reversible) rather than silently retained here.

11.6. Evolution Generation: Monotone Realization of Branch Time by the Licensed Generator. This subsection discharges O_Ham.Time (Rmk. 11.18). It is the first theorem of the corpus binding two licensed variables *dynamically*: the generator of the licensed energy form (Thm. 11.13) and the licensed continuation order (Thm. 11.23). The strategy is order-level throughout: “evolution” means motion along $\preceq_t$, and no smooth-time notation is used or licensed.

DEFINITION 11.26 ([D]). (H_B -Compatible Steps and the Generated Suborder.) Let B carry a licensed energy form $\mathfrak{h}_B$ (Thm. 11.9) with generator H_B (Thm. 11.13), a licensed continuation order $\preceq_t$ (Thm. 11.23), and a declared dissipation ledger on continuation steps. An admissible continuation step $x \rightarrow y$ is H_B -compatible if

- (i) it is a generating step of $\preceq_t$ (admissible and a certified quotient-visible state change); and
- (ii) it is *form-non-increasing*: $\mathfrak{h}_B(y) \leq \mathfrak{h}_B(x)$, with the decrement $\mathfrak{h}_B(x) - \mathfrak{h}_B(y)$ recorded on the declared dissipation ledger.

The H_B -generated suborder $\preceq_H$ is the preorder generated by H_B -compatible steps. By construction $\preceq_H \subseteq \preceq_t$.

LEMMA 11.27 ([C]). (Energy as Lyapunov Ledger along H_B -Compatible Continuation.) [Conditional on: Thm. 11.9, Thm. 11.13, and SCT.1 (**thm:SCT1**, Book II) on the branch.] *Along any $\preceq_H$ -chain the coercive cost $\mathfrak{h}_B$ is non-increasing, every decrement is ledger-visible, and the total cost dissipated along the chain equals the sum of the per-step ledger entries. Moreover, the canonical contraction family of the nonnegative generator is form-non-increasing: for every $s \geq 0$ (an internal spectral-calculus parameter, not a licensed time variable),*

$$\overline{\mathfrak{h}_B}(e^{-sH_B}\psi) \leq \overline{\mathfrak{h}_B}(\psi) \quad \text{for all } \psi \in \text{dom}(\overline{\mathfrak{h}_B}),$$

so each application of the contraction family whose quotient image is a certified state change is an H_B -compatible step.

PROOF. Monotonicity along a $\preceq_H$ -chain is Def. 11.26(ii) applied stepwise; the telescoping identity gives the total-dissipation accounting. Ledger-visibility of every decrement is SCT.1: all transport degradation is ledger-visible, and a decrement off the declared ledger would be hidden loss, which SCT.1 excludes; the reverse direction (no spontaneous coercive gain along admissible steps) is transfer-coercivity (Def. 2.10). For the contraction family: $H_B \geq 0$ is self-adjoint (Thm. 11.13), so by the spectral theorem $\overline{\mathfrak{h}}_B(e^{-sH_B}\psi) = \int_{[0,\infty)} \lambda e^{-2s\lambda} d\|E_\lambda\psi\|^2 \leq \int_{[0,\infty)} \lambda d\|E_\lambda\psi\|^2 = \overline{\mathfrak{h}}_B(\psi)$, since $\lambda e^{-2s\lambda} \leq \lambda$ on $\text{Spec}(H_B) \subseteq [0, \infty)$. The parameter s enters only through functional calculus of the licensed operator; no time-evolution claim is made or needed. $\square$ $\square$

THEOREM 11.28 ([C]). (Monotone Realization: When H_B -Dynamics Realizes Branch Time.) [Conditional on: Thm. 11.23, Thm. 11.13, Lem. 11.27, SCT.1 on the branch, and the declared dissipation-ledger discipline.]

- (1) (Order-compatibility.) *The H_B -generated family is monotone for branch time: every H_B -compatible step is a generating step of $\preceq_t$, hence $\preceq_H \subseteq \preceq_t$ and motion under the H_B -generated family never contradicts the licensed continuation order.*
- (2) (Realization criterion.) *H_B realizes branch time on a declared sector $S \subseteq \Sigma_B$ if and only if every generating step of $\preceq_t$ within S is H_B -compatible up to ledger-visible defect — equivalently, $\preceq_H|_S = \preceq_t|_S$. In that case “the evolution generated by H_B ” has licensed meaning on S as motion along the branch continuation order, and the demand of $O_Ham.Time$ is met on S .*
- (3) (Failure diagnosis.) *If $\preceq_H|_S \subsetneq \preceq_t|_S$, the discrepancy is ledger-visible: there exists a certified continuation step within S whose coercive-cost change is not accounted by the H_B dissipation ledger, witnessing either an energy channel outside the declared form (an undeclared hidden supplement, prohibited) or a continuation step imported from outside the admissible family (an imported flow). The failure of realization is therefore always nameable, never silent.*

Scope clause: *this theorem operates at the order level only. e^{-itH_B} with $t \in \mathbb{R}$ as licensed time remains unlicensed pending $O_Time.Encode$ clause (b) (Rmk. 11.25).*

PROOF. (1) By Def. 11.26(i), every H_B -compatible step is a generating step of $\preceq_t$; monotonicity of the generated preorder follows by induction on chain length, exactly as in the transport-law clause of Thm. 11.23. By Lem. 11.27 the canonical contraction family of H_B supplies form-non-increasing steps, so the H_B -generated family moves only along $\preceq_H \subseteq \preceq_t$.

(2) ($\Leftarrow$) Suppose every generating step of $\preceq_t$ within S is H_B -compatible up to ledger-visible defect. Then every generating chain of $\preceq_t|_S$ is a chain of H_B -compatible steps with all defects ledgered, hence a $\preceq_H$ -chain; so $\preceq_t|_S \subseteq \preceq_H|_S$, and with (1) the two preorders coincide on S . ($\Rightarrow$) Conversely, if $\preceq_H|_S = \preceq_t|_S$ then in particular each generating step $x \rightarrow y$ of $\preceq_t$ within S satisfies $x \preceq_H y$, i.e. it is reachable by H_B -compatible steps with ledgered accounting — which is the criterion. Under the criterion, “evolution” is motion along $\preceq_t$ realized by H_B -compatible steps; since $\preceq_t$ is the licensed time variable (Thm. 11.23) and H_B the licensed generator (Thm. 11.13),

the phrase “evolution generated by H_B ” carries exactly the licensed content and no more.

(3) Suppose $x \preceq_t y$ within S but $x \not\preceq_H y$. Every generating chain from x to y then contains a step $x' \rightarrow y'$ that fails Def. 11.26(ii) modulo the ledger: either $\mathfrak{h}_B(y') > \mathfrak{h}_B(x')$ with the increment unledgered, or the decrement disagrees with the ledger entry. By SCT.1 all lawful transport degradation is ledger-visible, so an unledgered coercive change cannot arise from the declared transport-coercion structure; it therefore witnesses either coercive content entering through a channel outside the declared form (a hidden supplement, prohibited by the standing no-hidden-supplementation discipline) or a step that is not generated by the admissible continuation family at all (an imported flow). Both diagnoses are recorded objects: the discrepancy set is ledger-visible by construction. $\square$ $\square$

REMARK 11.29 ([R]). (Realization Instances: NS Dissipative Discipline, YM Phase-A Persistence.) Two certified canonical regimes satisfy the realization criterion of Thm. 11.28(2) on their declared sectors. *NS*: on the safe-tail regime the contraction $\mathfrak{D}_{n+1} \leq \kappa \mathfrak{D}_n$ (`cor:NS61-discharged`, NS Branch) makes every admissible continuation step strictly form-non-increasing with the decrement carried on the obstruction ledger, so every $\preceq_t$ -generating step is H_B -compatible and NS realizes branch time on the declared sector. *YM*: on the Phase-A screened sector the depth-indexed persistence machinery (`thm:0-GLOB`, YM Branch) certifies each refinement stage as a quotient-visible change while the contraction family of the Friedrichs generator of `thm:B1-closable` is form-non-increasing (Lem. 11.27); the Phase-A sector is a realization sector. These are realization instances of an already-proved criterion; this remark confers no new force.

11.7. The Entropy Clock: Canonical Cross-Chain Comparison. The promotion of the foundation rung pays an immediate dividend on the temporal rung. Rmk. 11.24 narrowed `O_Time.Clock` to cross-chain canonicity of step counts; the Book I ledger machinery shows that raw step count was the wrong clock candidate all along, and supplies the right one.

THEOREM 11.30 ([C]). (Entropy Clock: Canonical Cross-Chain Comparison via the Defect Ledger.) [Conditional on: Thm. 11.2, Thm. 11.23; the Book I duality and preorder machinery (Thm. ??, Prop. ??, Prop. ??) for the spine instance; and the branch carrying a declared defect ledger Δ_B compatible with its admissible continuation steps.]

- (1) (Canonicity.) *Cumulative defect weight Δ_B is a quotient-visible, reparametrization-invariant chain functional: its value along a continuation chain depends only on the certified class-splitting and coarsening events the chain traverses, not on the presentation or step-partition of the chain. By Ledger–Entropy Duality, the cumulative defect along a chain equals the entropy decrement, so the entropy ledger and the defect ledger define the same clock.*
- (2) (Order-compatibility.) *The Δ_B -clock is monotone for branch time: along any admissible continuation Δ_B is non-decreasing, so $x \preceq_t y$ implies that every generating chain from x to y carries Δ_B -weight ≥ 0 , and the clock never runs backward against the licensed order.*

- (3) (Cross-chain synchronization; `O_Time.Encode` clause (a) conditionally discharged.) *Two continuation chains are lawfully synchronized exactly when their cumulative defect weights agree at corresponding certified events. Equal raw step count is neither necessary nor sufficient: chain depth is a lawful clock precisely on families of chains with uniform per-step defect, and otherwise is a presentation index. The lawful clock demanded by `O_Time.Encode` clause (a) is the entropy/defect ledger.*

PROOF. (1) Each entry of Δ_B records a certified class-splitting or coarsening event — a quotient-visible datum by construction (the defect ledger is the transaction log of the quotient, Def. ??, Book I). Re-partitioning a chain into different steps re-groups the event multiset but does not alter it; by ledger additivity (Prop. ??) the cumulative weight is the sum over the event multiset, hence partition-independent and re-description-invariant. The identity with the entropy decrement is Ledger–Entropy Duality (Thm. ??) on spine survivor chains, carried to branch continuation chains by clause (S3) of the licensed branch ledger (Thm. 11.2).

(2) Each admissible continuation step appends a nonnegative one-step defect (Prop. ??); induction on chain length gives monotonicity along every generating chain of $\preceq_t$, which is order-compatibility in the same sense as Thm. 11.28(1).

(3) ($\Leftarrow$) If two chains agree in cumulative defect at corresponding certified events, then by (1) they agree in every quotient-visible clock reading, and by duality in entropy decrement; no admissible probe can distinguish their temporal content, which is lawful synchronization. ($\Rightarrow$) Conversely, a disagreement in cumulative defect at corresponding events is itself a certified, quotient-visible discrepancy, so the chains are not synchronized. For step count: on a family of chains with uniform per-step defect δ_0 , depth n and weight $n\delta_0$ are affinely equivalent, so depth is lawful there; in general two chains of equal depth may traverse different event multisets, and (1) shows the weight, not the count, is the invariant. This is exactly the canonicity demanded by `O_Time.Encode` clause (a), conditionally discharged with the clock identified. $\square$ $\square$

REMARK 11.31 ([R]). (The Clock and the Arrow Are One Object.) The Book I irreversible temporal preorder (Sch. ??: $h \preceq h'$ iff $\Delta(h) \leq \Delta(h')$, forward-only by defect nonnegativity) is, after this subsection, recognizable as the spine instance of the entropy clock read as an order: the arrow of time and the clock of time are the same ledger viewed as order and as scale. No new force; this remark records the identification for the encoding work that remains.

11.8. The Canonical Nontrivial Sector: `O_Mass.Sector` Discharged by Three-Variable Binding. The sector hypothesis of Thm. 11.15 is the last piece of that theorem still entering as branch-supplied data. This subsection recovers it. The construction binds *three* licensed variables — the entropy ledger, the continuation order, and the generator — and its shape is forced by an audit failure recorded inside the theorem itself.

DEFINITION 11.32 ([D]). (Persistent, Label, and Nontrivial Sectors.) Let B carry the universal entropy ledger S_B (Thm. 11.4), the licensed continuation order $\preceq_t$ (Thm. 11.23), and the licensed generator H_B in its declared Hilbert realization (Thm. 11.13). Define:

- (i) the *persistent-distinction classes*

$$\Sigma_B^{\text{pers}} := \{x \in \Sigma_B : S_B(y) > 0 \text{ for every } y \succeq_t x\},$$

the classes from which every admissible future sustains at least one admissible distinction (note Σ_B^{pers} is up-closed for $\preceq_t$ by construction);

- (ii) the *persistent sector* $\mathcal{H}_B^{\text{pers}}$: the closed span, in the declared realization, of the states realizing classes of Σ_B^{pers} ;
 (iii) the *label sector* $\mathcal{H}_B^{\text{label}} := \mathcal{H}_B^{\text{pers}} \cap \ker H_B$: persistent distinctions of zero coercive cost;
 (iv) the *canonical nontrivial sector*

$$\mathcal{H}_B^{\text{nontriv}} := \mathcal{H}_B^{\text{pers}} \ominus \mathcal{H}_B^{\text{label}},$$

persistent distinctions of positive coercive cost.

Here $\ker H_B = \{\psi : \overline{\mathfrak{h}}_B(\psi) = 0\}$ is the kernel of the licensed closed form (for a nonnegative self-adjoint generator, $\overline{\mathfrak{h}}_B(\psi) = \|H_B^{1/2}\psi\|^2$, so the two kernels coincide).

THEOREM 11.33 ([C]). (O_Mass.Sector Discharge: the Canonical Nontrivial Sector.) [Conditional on: Thm. 11.4, Thm. 11.23, Thm. 11.13; and sector-realization compatibility: the declared realization carries the $\preceq_t$ -forward-invariance of Σ_B^{pers} to invariance of $\mathcal{H}_B^{\text{pers}}$ under the contraction family of H_B — automatic on realization sectors in the sense of Thm. 11.28(2).]

- (1) (Quotient-visibility; no branch-supplied data.) $\mathcal{H}_B^{\text{nontriv}}$ is determined by the three licensed variables and the declared realization alone: admissible re-description fixes S_B , $\preceq_t$, and $\overline{\mathfrak{h}}_B$ (hence $\ker H_B$), and therefore fixes the sector. No sector datum is imported.
- (2) (Reduction; the spectral threshold is well-posed.) $\mathcal{H}_B^{\text{pers}}$ reduces H_B , hence so does $\mathcal{H}_B^{\text{nontriv}}$, and $\inf \text{Spec}(H_B|_{\mathcal{H}_B^{\text{nontriv}}})$ is well-defined. Substituting this sector into Thm. 11.15 turns its sector hypothesis into recovered structure; only the positivity of the threshold remains branch data, by design.
- (3) (Coercivity audit, passed by forced factorization.) The naive candidate $\mathcal{H}_B^{\text{pers}}$ alone fails the audit: it contains persistent distinctions of zero coercive cost, on which the spectral threshold is 0, conflating distinguishability with excitation — the same gap that blocked the entropy-to-energy reduction (Rmk. 11.7). The factorization through $\ker H_B$ is therefore forced; it is licensed because the kernel of a licensed closed form is canonical.
- (4) (Byproduct: the label sector.) $\mathcal{H}_B^{\text{label}}$ is a derived canonical object: persistent, distinguishable, costless labels, stable under the H_B -generated family. Its phenomenology matches superselection-type structure; the identification with branch superselection sectors is recorded as an observation, not claimed as a theorem.

PROOF. (1) Each ingredient is quotient-determined: S_B by Thm. 11.4(2), $\preceq_t$ by Thm. 11.23 clause (2), and $\overline{\mathfrak{h}}_B$ by Thm. 11.9 clause (2) with its kernel determined by the form. The set Σ_B^{pers} is defined by a condition in S_B and $\preceq_t$ only; its span, the kernel intersection, and the orthogonal complement are determined by these data within the

declared realization. An admissible re-description fixes all three variables, hence the sector.

(2) Σ_B^{pers} is up-closed for $\preceq_t$: if every admissible future of x has positive burden, the same holds for every $y \succeq_t x$, whose futures are among the futures of x . By Thm. 11.28(1) the H_B -generated family moves only along $\preceq_t$, so at the quotient level it preserves Σ_B^{pers} ; by the sector–realization compatibility hypothesis this lifts to $e^{-sH_B} \mathcal{H}_B^{\text{pers}} \subseteq \mathcal{H}_B^{\text{pers}}$ for all $s \geq 0$. Since each e^{-sH_B} is self-adjoint, invariance of the closed subspace $\mathcal{H}_B^{\text{pers}}$ implies invariance of its orthogonal complement, so $\mathcal{H}_B^{\text{pers}}$ reduces every e^{-sH_B} and hence reduces H_B . The kernel $\ker H_B$ is the fixed-point space of the contraction family and reduces H_B trivially; intersections and orthogonal differences of reducing subspaces reduce, so $\mathcal{H}_B^{\text{nontriv}}$ reduces H_B and the restricted spectrum is well-defined. The substitution claim is immediate: Thm. 11.15 required a quotient-visible sector; clause (1) supplies it with no branch input, leaving $m_B^2 > 0$ as the sole branch-data hypothesis — which that theorem already declared as such.

(3) On $\mathcal{H}_B^{\text{label}} = \mathcal{H}_B^{\text{pers}} \cap \ker H_B$ the form vanishes, so $0 \in \text{Spec}(H_B|_{\mathcal{H}_B^{\text{pers}}})$ whenever the label sector is nonzero, and the threshold on the unfactored candidate is 0 regardless of any gap above the kernel. Distinguishability ($S_B > 0$, class-counting content) does not imply coercive cost ($\mathfrak{h}_B > 0$, transport content); the audit gap is exactly the one isolated in Rmk. 11.7, and the factorization removes precisely the zero-cost part and nothing else.

(4) Stability of the label sector under the H_B -generated family holds because $\ker H_B$ is the fixed-point space and $\mathcal{H}_B^{\text{pers}}$ is invariant by (2); persistence and distinguishability hold by membership in Σ_B^{pers} . No further claim is made. $\square$ $\square$

REMARK 11.34 ([R]). (Instance Verification: YM Phase-A.) The construction recovers the certified instance: on the YM Phase-A screened sector, persistence is supplied by the depth-indexed persistence machinery (`thm:0-GLOB`, YM Branch), the screening removes the zero-cost part, and the positive threshold is the certified gap $\Delta_{\text{YM}} = m^2 > 0$ — so the Phase-A screened sector instantiates $\mathcal{H}_B^{\text{nontriv}}$ with the realization-sector hypothesis certified at L+27 (Rmk. 11.29). NS realizes the construction trivially in the complementary sense: its declared regime concerns dissipation rather than a mass gap, and no gap claim is made. This remark records instances; no new force.

11.9. The Canonical Domain: O_Ham.Domain Discharged by Contraction Regularization. The last operator-theoretic gap in the Hamiltonian rung is the domain: Thm. 11.13 supplies $\text{dom}(\overline{\mathfrak{h}_B})$ as the form-norm completion of the declared core, but the obligation demands that this domain be *recovered* from finite probe and refinement data rather than presentation-supplied. The naive statement — “the domain is the closure of the finite-probe core” — would be a tautology if the core property were hypothesized; the content of this subsection is that the core property can be *derived*, because the licensed contraction family regularizes finite-probe data into a form core.

DEFINITION 11.35 ([D]). (Finite-Probe States; the Regularized Core.) Let B satisfy the hypotheses of Thm. 11.13 with declared Hilbert realization $\mathcal{H}_B$. A *finite-probe state*

is a state of $\mathcal{H}_B$ realizing a quotient class certified at finite refinement depth; write $\mathcal{D}_B^{\text{fin}}$ for their linear span. The *contraction-regularized finite-probe core* is

$$\mathcal{D}_B^{\text{reg}} := \text{span}\{e^{-sH_B}\psi : \psi \in \mathcal{D}_B^{\text{fin}}, s \geq 0\},$$

where the contraction family is the licensed functional calculus of the generator (Lem. 11.27: s is an internal spectral parameter, not a licensed time variable).

THEOREM 11.36 ([C]). (O_Ham.Domain Discharge: the Domain as Recovered Structure.) [Conditional on: Thm. 11.13; *realization faithfulness* — the finite-probe states are norm-dense in $\mathcal{H}_B$, which is the Hilbert-level expression of the no-hidden-channel discipline (`def:hidden-channel`, Book V): a state orthogonal to every finite-depth-certified state is invisible to every admissible probe, i.e. an undeclared channel of the realization; and *finite per-stage cost* — every finite-probe state lies in $\text{dom}(\mathfrak{h}_B)$.]

- (1) (Canonical core.) $\mathcal{D}_B^{\text{reg}}$ is a form core: $\text{dom}(\overline{\mathfrak{h}_B})$ is the closure of $\mathcal{D}_B^{\text{reg}}$ in the form norm $\|\psi\|_{\mathfrak{h}}^2 = \|\psi\|^2 + \overline{\mathfrak{h}_B}(\psi)$.
- (2) (Quotient-visible membership criterion.) $\psi \in \text{dom}(\overline{\mathfrak{h}_B})$ if and only if ψ is the norm limit of a form-Cauchy sequence from $\mathcal{D}_B^{\text{reg}}$ — a criterion expressed entirely in licensed data: finite-probe certificates, the licensed contraction family, and the licensed form. The domain is recovered structure, not presentation-supplied.
- (3) (Presentation independence.) Any two admissible finite-probe presentations that are mutually certifiable (each state of one is norm-approximable by certified states of the other — the admissible re-description discipline) yield regularized cores with the same form-norm closure.
- (4) (Finality.) Closedness of $\overline{\mathfrak{h}_B}$ makes the characterization final: no admissible refinement enlarges the domain beyond this closure, matching the discharge-by-construction of O_Ham.SA in Thm. 11.13.

Anti-smuggling audit: no Sobolev scale, manifold smooth structure, or a-priori operator domain is imported; the domain descends from probe certificates and licensed functional calculus alone.

PROOF. (1) *Density in $\mathcal{H}_B$:* by strong continuity of the contraction family of a nonnegative self-adjoint generator, $e^{-sH_B}\psi \rightarrow \psi$ as $s \rightarrow 0^+$; with realization faithfulness, $\mathcal{D}_B^{\text{reg}} \supseteq \mathcal{D}_B^{\text{fin}}$ is norm-dense. *Containment:* for $s > 0$, $e^{-sH_B}\psi \in \text{dom}(H_B^k)$ for every k ; for $s = 0$ the finite per-stage cost hypothesis gives $\psi \in \text{dom}(\mathfrak{h}_B)$. Hence $\mathcal{D}_B^{\text{reg}} \subseteq \text{dom}(\overline{\mathfrak{h}_B})$. *Invariance:* $e^{-tH_B}e^{-sH_B}\psi = e^{-(s+t)H_B}\psi \in \mathcal{D}_B^{\text{reg}}$.

It remains to show every $\psi \in \text{dom}(\overline{\mathfrak{h}_B})$ is form-approximable from $\mathcal{D}_B^{\text{reg}}$. Two spectral estimates suffice. *First*, for $\psi \in \text{dom}(\overline{\mathfrak{h}_B})$,

$$\|e^{-sH_B}\psi - \psi\|_{\mathfrak{h}}^2 = \int_{[0,\infty)} (1+\lambda) |e^{-s\lambda} - 1|^2 d\|E_\lambda\psi\|^2 \xrightarrow{s \rightarrow 0^+} 0$$

by dominated convergence: the integrand is dominated by $4(1+\lambda)$, integrable since $\psi \in \text{dom}(\overline{\mathfrak{h}_B})$, and tends to 0 pointwise. So it suffices to form-approximate $e^{-sH_B}\psi$ for

fixed $s > 0$. *Second*, the smoothing bound: for fixed $s > 0$ and any $\phi \in \mathcal{H}_B$,

$$\|e^{-sH_B}\phi\|_{\mathfrak{h}}^2 = \int_{[0,\infty)} (1+\lambda) e^{-2s\lambda} d\|E_\lambda\phi\|^2 \leq C_s \|\phi\|^2, \quad C_s := \sup_{\lambda \geq 0} (1+\lambda)e^{-2s\lambda} < \infty,$$

i.e. the contraction family is bounded from the Hilbert norm to the form norm. Now take finite-probe $\psi_n \rightarrow \psi$ in norm (faithfulness); then $e^{-sH_B}\psi_n \in \mathcal{D}_B^{\text{reg}}$ and $\|e^{-sH_B}(\psi_n - \psi)\|_{\mathfrak{h}}^2 \leq C_s \|\psi_n - \psi\|^2 \rightarrow 0$. Composing the two approximations gives a form-norm approximation of ψ from $\mathcal{D}_B^{\text{reg}}$, which is the core property; the form-norm closure of a core is $\text{dom}(\overline{\mathfrak{h}_B})$ by definition of the closure.

(2) Restates (1) through the definition of form-norm closure. Every ingredient of the criterion is licensed: the certificates are finite refinement data, the contraction family is licensed functional calculus of the licensed generator (Lem. 11.27), and the form is licensed (Thm. 11.9).

(3) Mutual certifiability gives mutual norm-density of the two finite-probe spans in each other's closures, hence both are norm-dense in $\mathcal{H}_B$; the smoothing bound then shows each regularized core form-approximates the other, so both closures equal $\text{dom}(\overline{\mathfrak{h}_B})$ by (1).

(4) The closure of a closable form is closed, and the Friedrichs domain is uniquely determined by the form (uniqueness clause of Thm. 11.13); any admissible refinement acts within the same licensed form and cannot pass beyond its closure. $\square$ $\square$

REMARK 11.37 ([R]). (Faithfulness Is Not an Import.) The realization-faithfulness hypothesis deserves its own record: it is not an analytic convenience but the Hilbert-level face of Book V lawfulness. A nonzero vector orthogonal to every finite-probe state would carry observational content reachable by no admissible probe at any finite depth — precisely an undeclared dependence of the realization on structure invisible through the declared projection, i.e. a hidden channel (`def:hidden-channel`, Book V). Lawful branches exclude it by discipline, not by assumption of convenience. No force conferred.

11.10. Linearization of Branch Time: O_Time.Encode Clause (b) by Characterization, and the Reversibility Obstruction. The clock theorem (§ 11.7) identified the canonical cross-chain comparison; what remained of O_Time.Encode was the order-to-number encoding itself. This subsection settles the discrete encoding by a full characterization (not merely a sectoral instance), supplies the continuum encoding as a bridge clause at the same epistemic level as the LSC clause of the energy form, and isolates — as a new, sharply named obligation — the one question the entropy clock provably cannot answer.

DEFINITION 11.38 ([D]). (Δ -Coherence; Clock Coordinate; Instants; Clock-Simultaneity.) Let B carry the licensed continuation order $\preceq_t$ (Thm. 11.23) and the canonical clock (Thm. 11.30). A $\preceq_t$ -connected subset $T \subseteq \Sigma_B$ with a declared origin o is Δ -coherent if any two generating chains between the same endpoints within T carry equal cumulative defect weight. On a coherent sector define the *clock coordinate*

$$\tau(x) := \Delta_B(\text{any generating chain } o \rightsquigarrow x),$$

well-defined by coherence. Call $x \sim y$ (*instant-equivalent*) when x and y are connected by a chain of zero-defect steps, and call x, y *clock-simultaneous* when $\tau(x) = \tau(y)$. A *uniform-defect family* is a family of chains of constant per-step defect $\delta_0 > 0$.

THEOREM 11.39 ([C]). (Linearization by Characterization: the Discrete Encoding of Branch Time.) [Conditional on: Thm. 11.23, Thm. 11.30.]

- (1) (Characterization.) *A canonical clock coordinate (a quotient-visible state function assigning every generating step its ledgered weight) exists on a $\preceq_t$ -connected sector T if and only if T is Δ -coherent. Necessity is immediate: path-dependent weight is inconsistent with any state function reproducing per-step ledger entries; sufficiency is Def. 11.38.*
- (2) (Affine gauge uniqueness.) *On a coherent sector, τ is monotone for $\preceq_t$, quotient-visible, and unique up to the clock gauge: any monotone encoding τ' assigning each generating step its ledgered weight in a declared unit satisfies $\tau' = a\tau + b$ (a the unit ratio, b the origin shift), by telescoping along any chain and coherence. The licensed time coordinate carries exactly the gauge freedom of a physical clock — unit and origin — and no more.*
- (3) (Strict monotonicity modulo instants; the fibers.) *For comparable $x \prec_t y$: $\tau(x) = \tau(y)$ iff $x \sim y$ (every connecting step is zero-defect), and otherwise $\tau(x) < \tau(y)$. The fibers of τ consist of instant-classes together with $\preceq_t$ -incomparable clock-simultaneous elements.*
- (4) (Canonical linear preorder; Szpilrajn avoided.) *The relation $x \leq_\tau y : \iff \tau(x) \leq \tau(y)$ is a linear preorder extending $\preceq_t$ on T , determined by licensed data plus gauge — no extension choice is made. A Szpilrajn-style linear order refining incomparable clock-simultaneous pairs would require exactly such a choice; canonicity is preserved by declining it: licensed time totally orders clock readings and deliberately does not refine simultaneity. Anti-smuggling: clock-simultaneity is a bookkeeping notion (equal ledgered weight), not a causal, metric, or signal-theoretic claim; cross-sector comparison remains governed by Thm. 11.30(3).*
- (5) (Uniform-defect specialization.) *On a uniform-defect family, $\tau = \delta_0 \cdot \text{depth}$: chain depth is an affine clock, recovering the criterion of Thm. 11.30(3) as the special case.*

PROOF. (1) If a state function τ' reproduces per-step ledger entries, then for any chain $o \rightsquigarrow x$ the telescoped sum gives $\tau'(x) - \tau'(o) = \Delta_B(\text{chain})$, so all chains to x carry equal weight: coherence is necessary. Conversely, on a coherent sector Def. 11.38 defines τ consistently.

(2) Monotonicity is defect-nonnegativity along chains (Thm. 11.30(2)); quotient-visibility is Thm. 11.30(1). For τ' as hypothesized, telescoping along any chain $o \rightsquigarrow x$ gives $\tau'(x) = \tau'(o) + a\Delta_B(\text{chain}) = b + a\tau(x)$, chain-independent by coherence.

(3) For $x \prec_t y$, every connecting chain has nonnegative per-step weights summing to $\tau(y) - \tau(x)$; the sum vanishes iff every step is zero-defect, i.e. $x \sim y$. The fiber description follows.

(4) $\leq_\tau$ is reflexive, transitive, and total (values in $\mathbb{R}$), and extends $\leq_t$ by (2)–(3). It is determined by τ , hence by licensed data and the gauge of (2). That no further canonical refinement exists is the content of (3): an incomparable clock-simultaneous pair is separated by no licensed datum, so any ordering of it is a choice — which the linear preorder declines by construction.

(5) Immediate: constant per-step defect δ_0 makes every chain weight $\delta_0 n$ at depth n . $\square$ $\square$

THEOREM 11.40 ([C]). (Continuum Encoding: the Bridge Clause.) [Conditional on: Thm. 11.39 on each sector; and a declared Book VI bridge presentation of the clock: a refining family $\{T_k\}$ of Δ -coherent sectors with admissible refinement maps under which the clock coordinates form a compatible monotone family, the value sets become ε_k -dense in a declared interval with $\varepsilon_k \rightarrow 0$, and the refinement defect of clock readings vanishes in the limit (the clock analogue of the LSC clause (E4)).] *There is a unique monotone continuous limit coordinate t on the bridge sector restricting compatibly to every τ_k up to gauge; continuum time notation $t \in \mathbb{R}$ is thereby licensed on the bridge sector for order and irreversible content, with regularity beyond continuity inherited from, and only from, the declared bridge regularity.*

PROOF. Compatibility and monotonicity give a well-defined monotone function on the union of value sets; ε_k -density with $\varepsilon_k \rightarrow 0$ makes this union dense in the declared interval, and a monotone function on a dense set with vanishing refinement defect extends uniquely to a continuous monotone function (the vanishing defect supplies the modulus of continuity, exactly as the LSC clause controls the energy form across the bridge). Uniqueness up to gauge follows from Thm. 11.39(2) applied at every stage. The scope restriction is part of the statement: nothing here licenses a parameter for defect-free dynamics — see Rmk. 11.41. $\square$ $\square$

REMARK 11.41 ([C]). **O_Time.Reversible — Conditionally Discharged by Dichotomy; the residual re-registered.** [Conditional on Thm. 11.42, Cor. 11.43, § 11.11.] The question as registered — whether a second clock for defect-free dynamics exists — is answered by the dichotomy: on the gauge horn no clock exists because no licensed change exists (e^{-itH_B} is admissible re-description); on the phase horn the clock exists and is constructed — cyclic/toral phase from licensed spectral differences, canonical up to origin + orientation gauge. On both horns, global linear unitary time is unlicensed, so thermodynamic and unitary time remain provably distinct. The genuinely open residual — canonical cross-orbit phase alignment, the defect-free analogue of clock comparison — is re-registered as Rmk. 11.45 (O_Phase.Sync), not retained here.

11.11. The Reversible-Clock Dichotomy: O_Time.Reversible Resolved. The L+32 obstruction asked whether any licensed datum can distinguish points of a unitary orbit — the prerequisite for any clock on defect-free dynamics. The feasibility analysis resolves into a dichotomy decidable from the declared probe family, with both horns in-toolkit; this subsection proves it, constructs the second clock on the horn where it exists, and shows that on *both* horns a global linear unitary time remains unlicensed — so the question splits correctly rather than closing falsely.

THEOREM 11.42 ([C]). (The Reversible-Clock Dichotomy.) [Conditional on: Thm. 11.13, Thm. 11.39; the declared admissible probe family of B ; and, for the phase-coordinate clause, orbit states of finite spectral support.] *Let I be an instant-class of a lawful branch and consider the orbits of the licensed unitary family $\{e^{-itH_B}\}$ within the realization over I . Exactly one of the following holds, and which one is determined by the declared probe family (whether some admissible probe fails to commute with the unitary family on I):*

- (I) (Gauge horn.) *Every admissible probe is constant along unitary orbits over I . Then orbit points lie in a single quotient class: the unitary family induces no certified change, acts as admissible re-description, and is pure presentation. A reversible clock does not exist because there is no licensed motion to measure; unitary “time” on I is unphysical in the NFC sense.*
- (II) (Phase horn.) *Some admissible probe separates orbit points. Then distinct orbit points realize distinct quotient classes, and the orbit motion is certified zero-defect change within the instant: each induced step is probe-witnessed, invertible (the family contains e^{+itH_B}), and class-count preserving, hence carries no ledger event. For orbit states of finite spectral support $\{\lambda_1, \dots, \lambda_r\}$, the certified orbit motion is periodic or quasi-periodic with frequency module generated by the licensed spectral differences $\lambda_i - \lambda_j$, and there exists a phase coordinate on each orbit — a toral coordinate of dimension the rank of the frequency module, cyclic in the rank-one case — canonical up to phase origin and orientation in each generator. The second clock exists, and it is cyclic/toral, not linear: its gauge group is origin + orientation per phase, in contrast to the affine unit + origin gauge of the thermodynamic clock (Thm. 11.39(2)).*

PROOF. The disjunction is exhaustive and exclusive by inspection of the declared probe family. **Horn (I):** the observational quotient is generated by admissible probe outcomes; if every outcome is constant along an orbit, all orbit points are outcome-indistinguishable, hence lie in one quotient class. A quotient-trivial action induces no certified state change, and a transformation inducing none is an admissible re-description by the definition of the quotient; there is no licensed change, so no clock has an object.

Horn (II): Let p be a separating probe. Orbit points with distinct p -outcomes realize distinct quotient classes, and each transition is certified by p itself. Zero defect: the induced steps occur at fixed refinement depth (no probe-family refinement is performed by the motion), are invertible within the licensed family, and preserve class counts; the defect ledger records class-splitting and coarsening events only (Thm. 11.30(1)), and the motion performs neither, so its ledger increment is zero and the entire orbit lies within one instant — consistent with, and required by, Thm. 11.39(3). For the phase coordinate: with finite spectral support, the spectral theorem gives

$$e^{-itH_B}\psi = \sum_{i=1}^r e^{-it\lambda_i} P_{\lambda_i}\psi,$$

so every probe outcome along the orbit is a function of the relative phases $e^{-it(\lambda_i - \lambda_j)}$ — the orbit’s certified image is a (quasi-)periodic trajectory whose frequency module is generated by the differences $\lambda_i - \lambda_j$, which are licensed data (differences of spectral values of the licensed generator, each a quotient-visible invariant in the sense of Thm. 11.15’s threshold clause). The map $t \mapsto (e^{-it(\lambda_i - \lambda_j)})_{i < j}$ descends to a coordinate on the closure of the orbit’s certified image: a torus of dimension the rank of the frequency module over $\mathbb{Z}$, a circle when the rank is one. Two such coordinates built from the same licensed frequency data differ by a choice of origin on each phase and an orientation of each generator — the stated gauge — by the same telescoping argument as Thm. 11.39(2), applied per frequency. $\square$ $\square$

COROLLARY 11.43 ([C]). (No Global Linear Unitary Time, on Either Horn.) [Conditional on Thm. 11.42.] *On neither horn is e^{-itH_B} with t a global linear licensed time variable licensed: on horn (I) because the family induces no licensed change at all, on horn (II) because the licensed parameter is cyclic/toral phase, and promoting phase to a global linear time would require canonically aligning phase coordinates across distinct orbits and instants — which is exactly the synchronization question that the entropy clock answers only for defect-bearing change (Thm. 11.30(3)) and that no licensed datum currently answers for defect-free change. e^{-itH_B} remains canon as internal functional calculus (Lem. 11.27 license note), now with its physical reading settled horn-by-horn.*

PROOF. Horn (I) is immediate. On horn (II), a global linear t would induce a simultaneous canonical choice of origin and orientation for the phase coordinate of every orbit; by the gauge clause of Thm. 11.42(II) no licensed datum on a single orbit makes this choice, and no cross-orbit licensed comparison exists for defect-free change — a canonical global choice would therefore be exactly the hidden selection that the licensing predicate’s declaration clause excludes. $\square$ $\square$

REMARK 11.44 ([R]). (Observation: Phase Alignment Is Connection-Type Data.) What horn (II) leaves open — a rule canonically aligning phase coordinates across nearby orbits and instants — has the formal shape of a *connection*: locally, alignment data on fibers of phases; globally, the obstruction to consistent alignment is holonomy-type. The corpus’s gauge-theoretic machinery (YM branch) is the natural eventual home of this observation, which would, if developed, recover gauge structure as the bookkeeping of reversible-clock synchronization. Recorded as an observation only; no force conferred, no development claimed.

REMARK 11.45 ([C]). **O.Phase.Sync (Cross-Orbit Phase Synchronization)** — **Conditionally Discharged by the Synchronization Obstruction.** [Conditional on Cor. 12.4, § 12.] Phase coordinates synchronize across an orbit family iff the phase holonomy vanishes on every independent cycle of the interface nerve — the obstruction in $H^1(N; T^r)$ supported on $\beta_1(N)$, with vanishing branch data and automatic when $\beta_1 = 0$. The expectation that this obligation is obstruction-bearing (recorded at surfacing) is confirmed at theorem level: the obstruction is the holonomy of the reversible-clock connection, and by Rmk. 12.5 it is supported on exactly the cycle rank that carries non-abelian gauge structure.

11.12. Characterization of the Energy Form: O_Energy.Form Resolved at the Ledger Level. The last in-toolkit VRP residual asked for universal existence of the branch energy form. The L+29 negative result (Rmk. 11.7) showed why no class-counting construction can deliver it; this subsection resolves the obligation the way the reversible-clock question resolved at L+33 — by characterization rather than one-sided answer. Existence at the ledger-functional level is equivalent to a declaration: a lawful branch carries an energy form exactly when it declares weighted transport-coercion structure, and on such branches the form is *constructed* canonically. The operator level is then carried by a hypothesis the corpus has named since L+24.

DEFINITION 11.46 ([D]). (Weighted Transport-Coercion Structure.) A lawful branch B carries *weighted transport-coercion structure* when the “declared structural data” of transfer coercivity (Def. 2.10) takes the quantitative form of *per-step coercion weights*: an assignment $w(\sigma) \geq 0$ to each admissible transfer step σ , declared with:

- (W1) (*Chain bound.*) The accumulated coercive change along a composable chain is bounded by the sum of its step weights — the quantitative instance of Def. 2.10;
- (W2) (*Coercivity gap.*) There is $c_B > 0$ such that every admissible chain from the transport-trivial class to a persistent nontrivial class accumulates total weight $\geq c_B$;
- (W3) (*Refinement stability.*) Weights are stable under admissible refinement: a refined step carries the weight of the step it refines, and refinement introduces no negative-weight steps.

THEOREM 11.47 ([C]). (O_Energy.Form Resolution: Characterization and Canonical Construction.) [Conditional on: Thm. 11.9; the Book V lawfulness discipline; and, for clause (2), the declared weighted structure of Def. 11.46.]

- (1) (Necessity, by extraction.) *If a lawful branch carries a branch energy form (Def. 11.8), then its declared data certifies weighted transport-coercion structure: the increments of $\mathfrak{h}_B$ along admissible steps are per-step weights satisfying (W1) (the form’s transport law), (W2) (clause (E2): positive cost of persistent nontrivial excitation), and (W3) (clause (E3)). An energy form is a weight declaration.*
- (2) (Sufficiency, by canonical construction.) *Conversely, given declared weighted structure, the canonical persistence cost*

$$\mathfrak{h}_B^{\min}(x) := \inf \left\{ \sum_{\sigma \in \gamma} w(\sigma) : \gamma \text{ an admissible transfer chain from the transport-trivial class to } x \right\}$$

satisfies clauses (E1)–(E3) of Def. 11.8; clause (E4) holds as the usual bridge clause on declared bridge regimes; and $\mathfrak{h}_B^{\min}$ is canonical: it is the maximal functional subordinate to the declared weights (any functional vanishing on the trivial class whose per-step increments respect the weights is dominated by it), determined by the declaration with no further choice.

- (3) (The residual is a pre-existing named hypothesis, not new open content.) *What the construction does not and cannot supply is clause (E5) at operator strength: a Hilbert realization in which $\mathfrak{h}_B^{\min}$ is a densely defined closable quadratic form.*

This is precisely the realization hypothesis carried, by name, in the bracket of Thm. 11.13 since its promotion; YM, GR, and NS are the certified models in which it holds. The universal-existence obligation therefore resolves with no new obligation surfaced: ledger-level existence is characterized and constructed; operator-level existence is conditional on declared realization, where it always was.

PROOF. (1) Set $w(\sigma) :=$ the ledgered increment of $\mathfrak{h}_B$ along σ (nonnegative in the gain direction by transfer-coercivity; decrements ride the dissipation ledger and bound the accumulated change). (W1) is the form's transport law (Thm. 11.9 clause (3)) telescoped along chains. (W2): a chain from trivial to a persistent nontrivial class x accumulates at least $\mathfrak{h}_B(x) > 0$ by clauses (E1)–(E2); take c_B the infimum of $\mathfrak{h}_B$ over persistent nontrivial classes, positive on the stabilized finite quotient (Thm. 11.4(1) supplies finiteness for lawful branches). (W3) is clause (E3).

(2) *Well-defined and* (E1): the trivial class is reachable from itself at cost 0, so $\mathfrak{h}_B^{\min}(\text{triv}) = 0$; for persistent nontrivial x , every admissible chain accumulates $\geq c_B$ by (W2), so $\mathfrak{h}_B^{\min}(x) \geq c_B > 0$ — which is also (E2), with the zero set exactly the transport-trivial (zero-cost-reachable) classes. (E3): under admissible refinement the chain family for a refined class consists of refinements of coarse chains (paths must respect the finer quotient), so the infimum is taken over a smaller constraint set with weights preserved ((W3)): $\mathfrak{h}_B^{\min}$ is non-decreasing under refinement — recorded cost is never spuriously destroyed. *Transport law*: chain concatenation gives the subordinate inequality $\mathfrak{h}_B^{\min}(y) \leq \mathfrak{h}_B^{\min}(x) + w(x \rightarrow y)$ — the no-spurious-gain face of transfer-coercivity, and the governing pattern (Thm. 9.1) in its coercive instance. *Maximality and canonicity*: if $\mathfrak{h}$ vanishes on the trivial class and its per-step increments respect the weights, then telescoping along any chain γ to x gives $\mathfrak{h}(x) \leq \sum_{\sigma \in \gamma} w(\sigma)$; taking the infimum over γ yields $\mathfrak{h} \leq \mathfrak{h}_B^{\min}$. The construction uses the declaration and nothing else. (E4): on declared bridge regimes, lower semicontinuity of a path-infimum cost under vanishing refinement defect is the same bridge argument as Thm. 11.40, applied to the weight ledger in place of the clock ledger.

(3) A path-infimum cost on a quotient is not, as such, a quadratic form on a Hilbert space; no licensed datum of the declaration selects a realization, and selecting one is Book V realization-functor territory (`def:legitimacy-witness`), declared per branch. The hypothesis is already named verbatim in the bracket of Thm. 11.13; nothing new is open. $\square$ $\square$

REMARK 11.48 ([R]). (The Question Deliberately Not Registered as an Obligation.) “When does the canonical persistence cost admit a closable quadratic realization?” — in effect, *why is energy quadratic?* — is recorded here at observation strength and deliberately *not* registered as an [O]. Reason: it is a realization-selection question (which Hilbert realization a branch declares, Book V functor territory), not a theorem target within the recovery program; registering it as an obligation would misclassify a declaration choice as an unproved claim. The certified models answer it instance-wise: in YM the declared weights are curvature increments and the canonical cost is realized

by the curvature form on its sector (`thm:B1-closable`, YM Branch), with GR and NS as the parallel instances of Rmk. 11.10. No force conferred.

11.13. Temperature as the Entropy–Energy Exchange Slope. With the entropy ledger universal (Thm. 11.4) and the energy form characterized (Thm. 11.47), the Temperature rung — anticipated at the gate note of Rmk. 11.3 — has both inputs canonical. Temperature is not a primitive ledger but a *response coefficient*: the marginal rate at which coercive cost must be spent to buy distinguishability along an admissible exchange family. The construction is choice-free given an exchange family, and the one classical ingredient it must refuse to import — the ensemble — is refused by construction, not by fiat: no partition function appears because the slope *is* the definition.

DEFINITION 11.49 ([D]). (Admissible Exchange Family; Exchange-Coherence.) Let B carry the universal entropy ledger S_B and a licensed energy form $\mathfrak{h}_B$ (its ledger reading E_B). An *admissible exchange family* on a sector $X \subseteq \Sigma_B$ is a set of admissible steps each of which changes both ledgers by recorded increments $(\Delta S, \Delta E)$ with $\Delta E \neq 0$, closed under the branch’s admissible composition. The family is *exchange-coherent* if the marginal ratio

$$\beta_B := \frac{\Delta S_B}{\Delta E_B}$$

is the same for all steps between any two states of X — the exchange analogue of Δ -coherence (Def. 11.38), i.e. path-independence of the entropy bought per unit coercive cost spent.

THEOREM 11.50 ([C]). (VRP-Temperature: the Exchange Slope as Licensed Response Coefficient.) [Conditional on: Thm. 11.4, Thm. 11.9 (with a licensed energy form on the sector); and an exchange-coherent admissible family (Def. 11.49).] *On an exchange-coherent sector the marginal slope $\beta_B = \partial S_B / \partial E_B$ is a licensed physics variable, realized through the certified branch-interface coefficient **disjunct of `def:variable-licensing`** (Book VII); where $\beta_B > 0$, the branch temperature is $T_B := \beta_B^{-1}$. The five recovery clauses:*

- (1) Probe signature: *the admissible exchange steps and the two ledger increments they record.*
- (2) Quotient invariant: *β_B is a ratio of two quotient-visible ledger increments (Thms. 11.4, 11.9), constant on the sector by exchange-coherence, hence invariant under admissible re-description.*
- (3) Transport law: *along lawful interfaces the slope transforms by the ratio of the two ledgers’ transfer behavior — entropy lax-functorially with no exchange coefficient (Thm. 11.5), energy with its declared exchange coefficient (Thm. 11.20) — so β_B carries exactly the inverse energy exchange coefficient and no independent transfer freedom.*
- (4) Coherence: *the slope is a state function precisely on exchange-coherent sectors; the discharge of the registered obligation `O_Temp.Slope` is this clause.*
- (5) Anti-smuggling audit (`O_Temp.Ensemble`, discharged by construction): *no canonical ensemble, partition function, heat bath, Boltzmann factor, or fitted β*

appears. β_B is the recorded marginal exchange ratio of two licensed ledgers; the ensemble is not refused by assumption but is simply never introduced, because the response slope is defined directly.

Sign clause (by design, not omission): *positivity $\beta_B > 0$ — coercive cost and distinguishability increasing together — is sector data, exactly as gap positivity is branch data in Thm. 11.15. Exchange families that trade the ledgers oppositely yield $\beta_B \leq 0$ (the population-inversion regime); the theorem licenses the slope at whatever sign the recorded increments give, and reserves the name “temperature” for the positive branch.*

PROOF. Quotient-visibility of each increment is Thm. 11.4(2) for S_B and Thm. 11.9 clause (2) for E_B ; their ratio is therefore quotient-determined wherever it is well-defined, and exchange-coherence (Def. 11.49) is exactly the condition that the ratio is a single sector constant — established by the same telescoping argument as Thm. 11.39(1): a path-dependent marginal ratio is inconsistent with any state function of the two ledgers, and on a coherent sector the common ratio defines β_B . The transport law is the quotient of the two transfer theorems: under a lawful interface ΔS_B is carried with no exchange coefficient (Thm. 11.5) and ΔE_B with coefficient g_I (Thm. 11.20), so $\beta_B \mapsto g_I^{-1}\beta_B$ — the certified branch-interface coefficient disjunct of the licensing predicate is met with β_B ’s sole transfer freedom being the inverse energy coefficient. The declaration clause is satisfied by the itemization in the statement, including the explicit sign clause and the anti-smuggling audit. No ensemble construct is used at any step. $\square$ $\square$

REMARK 11.51 ([R]). (Instance Pattern: the Dissipation Ledger Reads as a Temperature.) The L+27 dissipation ledger (Lem. 11.27) is the natural exchange family: along H_B -compatible continuation the coercive cost $\mathfrak{h}_B$ decreases by ledgered amounts, and where that spent cost registers as an entropy increase the recorded ratio $\Delta S_B/\Delta E_B$ is precisely β_B — a fluctuation–dissipation-shaped reading in which dissipated coercive cost and produced distinguishability are the two sides of one ledgered step. In the NS safe-tail regime the contraction $\mathfrak{D}_{n+1} \leq \kappa \mathfrak{D}_n$ (**cor:NS61-discharged**, NS Branch) supplies a sign-definite exchange family of exactly this shape. Recorded as an instance pattern; no force conferred and no equilibrium claimed.

REMARK 11.52 ([C]). **O_Temp.Equil (the Zeroth Law) — Conditionally Discharged by the Synchronization Obstruction.** [Conditional on Cor. 12.3, § 12.] The zeroth law holds as transitivity of thermal equilibrium along composable interfaces (by exchange-coefficient multiplicativity), and global temperature synchronization across a family is characterized: it holds iff the exchange-coefficient holonomy vanishes on every independent cycle of the interface nerve, the obstruction in $H^1(N; \mathbb{R}_{>0})$ supported on $\beta_1(N)$, automatic when $\beta_1 = 0$. Whether the obstruction vanishes for a given branch’s nerve is branch data, exactly as gap positivity is branch data (Thm. 11.15). The shell’s O_Temp.Kinetic (kinetic-theory reading) remains out of toolkit and is not registered as an in-corpus [O].

11.14. Charge as Licensed Conserved Quotient Invariant. The conserved charge space $\mathcal{Q}_\lambda = \ker(T_\partial - \lambda I)^\top$ (Def. 10.1) and its compatibility gate (Prop. 10.4,

unconditional) are already canonical. This subsection licenses the charge as a recovered physics variable, binding it to the continuation order (conservation) and to the type-transition operator (quantization), with the gate supplying the transfer law at no extra cost. Charge is the most directly NFC-native of the recovered variables after entropy: it is a left-kernel of an operator the canon already carries, not a new construction.

DEFINITION 11.53 ([D]). (Charge Observable.) For $\psi \in \mathcal{Q}_\lambda$ (Def. 10.1), the *charge observable* is the linear functional

$$q_\psi : \mathcal{O}_B \rightarrow \mathbb{R}, \quad q_\psi(x) := \langle \psi, x \rangle,$$

the left-eigenvector pairing of the observable state x with the conserved direction ψ . The *charge spectrum* is the set $\{\lambda\}$ of eigenvalues of T_∂ carrying a nonzero $\mathcal{Q}_\lambda$.

THEOREM 11.54 ([C]). (VRP-Charge: Conserved Charge as Licensed Quotient Invariant.) [Conditional on: the Book II type-transition structure (Def. ??) and Def. 10.1.] *The charge observable q_ψ (Def. 11.53) is a licensed physics variable in the sense of **def:variable-licensing** (Book VII), realized through the quotient-visible-invariant disjunct. The five recovery clauses:*

- (1) Probe signature: *the left-eigenvector pairing $\langle \psi, \cdot \rangle$, evaluated by the declared admissible probes that read the type-transition structure.*
- (2) Quotient invariant: *q_ψ is conserved under every admissible transfer map (Def. 10.1), so its value is constant on observational-equivalence classes and invariant under admissible re-description.*
- (3) Transport law: *supplied unconditionally by the compatibility gate (Prop. 10.4): q_ψ is the exact obstruction to interface solvability ($A_\lambda u = J_w$ solvable iff $q_\psi(J_w) = 0$ for all $\psi \in \mathcal{Q}_\lambda$), so charge transfers across lawful interfaces by the Fredholm pairing with no independent transfer freedom.*
- (4) Additivity: *q_ψ is linear, hence additive over independent (direct-sum) subsystems — the ledger property of a physical charge.*
- (5) Anti-smuggling audit: *no Noether theorem, Lagrangian, action principle, continuous symmetry group, or gauge potential is primitive; charge is the left-kernel of the licensed type-transition operator, recovered from canonical structure, not assumed from a symmetry postulate.*

PROOF. Conservation under admissible transfer is Def. 10.1 (left eigenvectors of T_∂ are fixed, up to the eigenvalue scaling, under the transfer maps generated by T_∂), giving clause (2): the pairing is constant on equivalence classes and re-description-invariant. Clause (3) is Prop. 10.4 verbatim — an unconditional result, so the transport law is inherited at full force. Linearity of $\langle \psi, \cdot \rangle$ gives clause (4). Clauses (1) and (5) are the declaration: the probe signature is the pairing, and the anti-smuggling audit holds because every ingredient (T_∂ , its left kernel, the pairing) is canonical NFC structure with no symmetry primitive introduced. The quotient-visible-invariant disjunct of the licensing predicate is thus satisfied. $\square$ $\square$

COROLLARY 11.55 ([C]). (Charge Is Conserved Along the Licensed Continuation Order.) [Conditional on Thm. 11.54, Thm. 11.23.] *For every $\psi \in \mathcal{Q}_\lambda$, the charge q_ψ is constant along the licensed continuation order $\preceq_t$: if $x \preceq_t y$ then $q_\psi(x) = q_\psi(y)$.*

Charge is thereby bound to branch time as its conserved companion — a further inter-variable binding (charge $\leftrightarrow$ order), alongside the generator–order binding (L+27) and the entropy–order binding (L+28).

PROOF. Each generating step of $\preceq_t$ (Def. 11.22) is an admissible quotient-visible state change, hence realized by an admissible transfer map, under which $\mathcal{Q}_\lambda$ is conserved (Def. 10.1). By clause (2) of Thm. 11.54 the pairing is unchanged across each step; induction on chain length gives constancy along $\preceq_t$. $\square$ $\square$

THEOREM 11.56 ([C]). (Charge Discreteness in the Finite Regime; Continuum Survival Open.) [Conditional on Thm. 11.54; finite-dimensional type space V at each refinement stage.] *On the finite NFC regime the charge spectrum $\{\lambda\}$ is discrete: T_∂ acts on a finite-dimensional type space, so its spectrum is finite and the admissible charge values are isolated. Charge quantization is therefore not an external postulate but a consequence of the finite type-transition structure — charges are discrete because types are. What is not established here is survival of discreteness under the Book VI continuum bridge: whether the isolated charge spectrum persists, or accumulates, in the continuum encoding is the open obligation `O.Charge.Quant-cont` (Rmk. 11.58).*

PROOF. A linear operator on a finite-dimensional space has finite spectrum; the eigenvalues carrying nonzero left-kernel $\mathcal{Q}_\lambda$ are among them, hence finite and isolated. Discreteness of the admissible charge values follows immediately. The continuum clause is explicitly deferred and proved nothing here. $\square$ $\square$

REMARK 11.57 ([R]). (Charge, the Gauge Template, and Cycle Rank.) The conserved charges organize the canonical gauge content: the YM gauge template $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ (`prop:YM-status`) and the SM matter quantum numbers are charge data in the present sense. By the synchronization obstruction (Thm. 12.2) the global organization of charges across an interface family is a class in $H^1(N; G)$ supported on $\beta_1(N)$, the same cycle rank that `thm:beta1-controls-nonabelian` (§13) assigns to non-abelian gauge structure. Charge, gauge structure, and synchronization holonomy thus share one support — the conserved charge is the local datum whose global gluing the cycle rank governs. Recorded as the program’s reading; no force conferred beyond the cited theorems.

REMARK 11.58 ([O]). **Open obligation `O.Charge.Quant-cont` (Continuum Survival of Charge Discreteness).** Establish whether the discrete charge spectrum of the finite regime (Thm. 11.56) survives the Book VI continuum bridge — i.e. whether admissible charge values remain isolated, or can accumulate, in the continuum encoding. This is the charge-rung instance of the standing continuum-encoding question (cf. the energy LSC clause and the time smooth-encoding clause); it is the continuum-bridge question for finite-regime spectral discreteness generally — it covers both the charge spectrum (Thm. 11.56) and the frequency module (Thm. 11.60, Rmk. 11.63), which present the identical deferral — the finite-regime discreteness being unconditional within its declared scope in both cases.

11.15. Frequency and Phase: Licensing the Spectral-Gap Module and Its Conjugate. The reversible-clock dichotomy (Thm. 11.42, phase horn) already

constructed, structurally, the phase coordinate on a unitary orbit and the frequency module generated by the licensed spectral gaps $\lambda_i - \lambda_j$. This subsection licenses *frequency* as a physics variable in its own right — the spectral-gap module of the licensed generator — establishes the phase–frequency duality (new: the phase coordinate of L+33 was built but never paired with frequency as conjugate), and binds frequency to the Hamiltonian spectrum, of which mass (Thm. 11.15) is the bottom gap. No dimensional constant and no energy–frequency physical law is introduced; the content is the dimensionless module structure and its conjugacy.

DEFINITION 11.59 ([D]). (Frequency Module.) On an orbit/sector I of finite spectral support $\{\lambda_1, \dots, \lambda_r\}$ of the licensed generator H_B (Thm. 11.13), the *frequency module* is the additive subgroup

$$\Omega_B(I) := \langle \lambda_i - \lambda_j : 1 \leq i, j \leq r \rangle \subseteq \mathbb{R},$$

generated by the spectral gaps. Its rank is the toral dimension of the phase coordinate of Thm. 11.42(II).

THEOREM 11.60 ([C]). (VRP-Frequency: the Spectral-Gap Module as Licensed Variable.) [Conditional on: Thm. 11.13, Thm. 11.15 (for the spectral-invariance of gaps); finite spectral support on the sector.] *The frequency module $\Omega_B(I)$ (Def. 11.59) is a licensed physics variable through the quotient-visible-invariant disjunct of **def:variable-licensing** (Book VII). The five recovery clauses:*

- (1) Probe signature: *the spectral gaps resolved by the admissible probes on the sector (the same probes that, on the phase horn, separate orbit points, Thm. 11.42(II)).*
- (2) Quotient invariant: *the gaps $\lambda_i - \lambda_j$ are differences of spectral values of the licensed generator, each a quotient-visible invariant (the spectral-threshold argument of Thm. 11.15 applied to every level, not only the bottom), so $\Omega_B(I)$ is fixed under admissible re-description.*
- (3) Transport law: *the generator transfers by uniqueness (Prop. 11.14), carrying its spectrum and hence the gap module; frequency transfers with the Hamiltonian and inherits no independent transfer freedom.*
- (4) Module structure (additivity): *composing admissible transitions adds their gaps, so $\Omega_B(I)$ is closed under addition — the ledger property, here a $\mathbb{Z}$ -module of frequencies.*
- (5) Anti-smuggling audit: *no Planck constant, no $E = \hbar\omega$, no oscillator, wave, or periodic-motion primitive, and no external clock; frequency is the spectral-gap structure of the licensed generator, recovered, not posited.*

PROOF. Clause (2) is the per-level spectral-invariance: each λ_i is a spectral value of the licensed self-adjoint H_B on a quotient-visible sector, invariant under admissible re-description by the same argument that makes the bottom gap m^2 invariant in Thm. 11.15; differences and the group they generate are then invariant. Clause (3): H_B is the unique generator of the transferred closed form (Prop. 11.14), so its spectrum, gaps, and gap module are carried by the same uniqueness. Clause (4) is closure of $\langle \lambda_i - \lambda_j \rangle$ under addition. Clauses (1),(5) are the declaration, the audit holding

because every ingredient is spectral data of the licensed generator with no dimensional or dynamical primitive introduced. The quotient-visible-invariant disjunct is satisfied. $\square$ $\square$

THEOREM 11.61 ([C]). (Phase–Frequency Duality.) [Conditional on Thm. 11.60, Thm. 11.42(II).] *On a phase-horn sector the phase coordinate is a torus T^r (Thm. 11.42(II)) and the frequency module $\Omega_B(I)$ is its character lattice: $\Omega_B(I) \cong \widehat{T^r} \cong \mathbb{Z}^r$, and phase and frequency are conjugate in the Pontryagin sense — each admissible probe outcome along the orbit is a finite combination of characters $e^{i\langle\omega,\theta\rangle}$, $\omega \in \Omega_B(I)$, $\theta \in T^r$. The duality pairing is the licensed, re-description-invariant pairing of the two licensed variables. Scope clause: this is abelian Pontryagin duality (torus $\leftrightarrow$ character lattice), not a quantum uncertainty relation and not an energy–frequency identification; no $\hbar$, no $\Delta\theta \Delta\omega$ bound, and no dimensional constant is asserted or licensed.*

PROOF. The phase coordinate descends to T^r with $r = \text{rank } \Omega_B(I)$ (Thm. 11.42(II)); the characters of T^r are indexed by $\mathbb{Z}^r$, which is exactly the frequency module under the gap generators (Def. 11.59). Each orbit observable is, by the spectral expansion $e^{-itH_B}\psi = \sum_i e^{-it\lambda_i} P_{\lambda_i}\psi$, a finite combination of the relative-phase characters $e^{-it(\lambda_i - \lambda_j)}$, i.e. a function on T^r with frequencies in $\Omega_B(I)$. Pontryagin duality of T^r and $\mathbb{Z}^r$ gives the conjugacy; both sides are licensed (phase by L+33, frequency by Thm. 11.60) and the pairing is re-description-invariant. The scope clause is a restriction on the claim, introducing nothing. $\square$ $\square$

COROLLARY 11.62 ([R]). (Frequency and Mass Are Two Readings of One Spectrum.) *Frequency is bound to the Hamiltonian as its gap structure: Ω_B is generated by differences of the spectrum whose bottom gap on the nontrivial sector is the mass m^2 (Thm. 11.15). Mass is the bottom gap; the frequency module is the full gap lattice; both are quotient-visible spectral invariants of the one licensed generator. This is the fifth inter-variable binding (frequency $\leftrightarrow$ generator/mass), and it records that the recovered spectral variables — mass and frequency — are not independent posits but two readings of $\text{Spec}(H_B)$. No force beyond the cited theorems.*

REMARK 11.63 ([R]). (Synchronization, Continuum Survival, and What Stays Out of Toolkit.) Three honest scope notes. (i) *Cross-orbit comparison* of phase and frequency across an interface family is the synchronization obstruction already characterized (Thm. 12.2, Cor. 12.4): a class in $H^1(N; T^r)$ on β_1 ; no new obligation. (ii) *Continuum survival* of the discrete frequency module is the frequency-rung instance of the same continuum-bridge question registered for charge (Rmk. 11.58, O_Charge.Quant-cont): on the finite regime Ω_B is discrete because spectral support is finite; whether discreteness survives the Book VI bridge is that one shared question, not a separate obligation. (iii) *Out of toolkit*: the physical energy–frequency law $E = \hbar\omega$ requires a dimensional quantization constant $\hbar$ that no licensed variable supplies; the in-toolkit content is the dimensionless module and its duality, and the $\hbar$ -identification is explicitly outside the recovery program.

12. Synchronization of Recovered Variables: One Obstruction, Supported on Cycle Rank

The recovery program leaves a single in-corpus frontier: two obligations — cross-orbit phase synchronization (`ob:vrp-phase-sync`, from the reversible-clock dichotomy) and the temperature zeroth law (`ob:vrp-temp-equil`) — that the surfacing passes independently flagged as connection-shaped. This section proves they are one obstruction. Each asks when a locally canonical quantity, defined up to a structure-group action and related across lawful interfaces by a multiplicative coefficient, glues into a single global quantity; the answer is cohomological, the obstruction lives in the first cohomology of the interface nerve, and — by the canon's own cycle-rank theorem (§ 13) — it is supported exactly on the complexity parameter β_1 that already governs non-abelianity. Synchronization is free precisely where the gauge structure is trivial.

DEFINITION 12.1 ([D]). (Synchronization Datum over an Interface Nerve.) Let $\{B_i\}$ be lawful sectors with lawful interfaces $I_{ij} : B_i \rightarrow B_j$ on overlaps, forming the *interface nerve* N (vertices the sectors, edges the lawful interfaces, triangles the certified triple overlaps). A *synchronization datum* valued in an abelian structure group G assigns:

- (i) to each sector a locally canonical invariant determined up to the action of G (its *gauge*: origin-and-orientation for a phase, multiplicative scale for an exchange slope);
- (ii) to each interface a transition element $g_{ij} \in G$ relating the two local invariants, with $g_{ji} = g_{ij}^{-1}$;

such that the transitions are *consistent on triple overlaps*: $g_{ij} g_{jk} g_{ki} = 1$ on every triangle of N (the *cocycle condition*). A datum is *synchronizable* when there exist gauge choices $a_i \in G$ with $g_{ij} = a_i a_j^{-1}$ for all edges — i.e. the local invariants glue, in a common gauge, into one global quantity.

THEOREM 12.2 ([C]). (The Synchronization Obstruction is a Cohomology Class Supported on β_1 .) [Conditional on: a synchronization datum (Def. 12.1) over a finite interface nerve N with abelian structure group G .]

- (1) (Cohomological characterization.) *The datum is synchronizable if and only if its transition family is a coboundary; equivalently, its class $[g] \in H^1(N; G)$ vanishes. This class is the complete obstruction: synchronization holds iff $[g] = 0$.*
- (2) (Support on cycle rank.) *For connected N , $H^1(N; G) \cong \text{Hom}(H_1(N), G)$, and $H_1(N)$ is free of rank $\beta_1(N)$. Hence the obstruction is carried by $\beta_1(N)$ independent generators — one holonomy per independent cycle of the interface nerve — and vanishes identically when $\beta_1(N) = 0$ (tree-like interface families): on a simply-connected nerve every consistent local datum synchronizes, with the common gauge fixed by transport along the spanning tree, uniquely up to one global G -choice.*
- (3) (Holonomy form.) *For each cycle $\gamma = (i_0 \rightarrow i_1 \rightarrow \cdots \rightarrow i_r \rightarrow i_0)$ of N the obstruction is the holonomy $\text{hol}(\gamma) = g_{i_0 i_1} g_{i_1 i_2} \cdots g_{i_r i_0} \in G$; the datum synchronizes iff $\text{hol}(\gamma) = 1$ on a generating set of cycles.*

PROOF. (1) If $g_{ij} = a_i a_j^{-1}$ then on any triangle the product telescopes to 1, so a coboundary is a cocycle (the datum is consistent) and is synchronizable by the gauges a_i . Conversely, synchronizability is by definition the existence of such a_i , i.e. that the cocycle g is a coboundary; the quotient of cocycles by coboundaries is $H^1(N; G)$ by definition, so synchronizability is exactly the vanishing of $[g]$.

(2) For a finite connected 1-complex (and, on triple overlaps, 2-complex) with abelian G , the universal-coefficient computation gives $H^1(N; G) \cong \text{Hom}(H_1(N), G)$; $H_1(N)$ of a finite complex is free abelian of rank the first Betti number $\beta_1(N)$. A homomorphism out of a free group of rank β_1 is determined by its values on β_1 generators, so the obstruction has exactly $\beta_1(N)$ independent components. When $\beta_1 = 0$ the group H_1 is trivial, $H^1 = 0$, and every cocycle is a coboundary: fix any spanning tree, set $a_{i_0} = 1$ at a root, and define a_i by the transition product along the unique tree path; consistency (the cocycle condition) makes this independent of how non-tree edges would route, so $g_{ij} = a_i a_j^{-1}$ on all edges.

(3) The holonomy around γ is the coboundary defect of the cycle; by (1)–(2) the class $[g]$ is determined by the holonomies on a generating set of $H_1(N)$, and $[g] = 0$ iff all such holonomies are trivial. $\square$ $\square$

COROLLARY 12.3 ([C]). (Zeroth Law and Temperature Synchronization.) [Conditional on Thm. 12.2; Thm. 11.50; and exchange-coefficient multiplicativity, Thm. 11.20(3).] Take $G = (\mathbb{R}_{>0}, \times)$ and transition elements the energy exchange coefficients $g_{ij} = g_{I_{ij}}$ relating the slopes β_{B_i} (Thm. 11.50(3)). Multiplicativity makes this a synchronization datum. Then:

- (1) (Zeroth law, transitivity.) *Along any path of composable interfaces, equality of g -corrected slopes is transitive — thermal equilibrium is an equivalence relation on path-connected sectors — by multiplicativity composing the corrections.*
- (2) (Global common temperature.) *A family of sectors admits a single common temperature (one global slope in a common gauge) iff the exchange-coefficient holonomy vanishes on every independent cycle of the interface nerve, i.e. iff the obstruction in $H^1(N; \mathbb{R}_{>0})$ vanishes; automatic when $\beta_1(N) = 0$.*

This conditionally discharges O-Temp.Equil: the zeroth law holds as transitivity, and global synchronization is characterized with the obstruction named; whether the obstruction vanishes for a given branch's interface nerve is branch data, exactly as gap positivity is branch data in Thm. 11.15.

PROOF. Multiplicativity (Thm. 11.20(3)) is the cocycle condition for $g_{ij} = g_{I_{ij}}$, so Def. 12.1 is met. (1) is the path-level specialization (composites multiply); (2) is Thm. 12.2(1)–(2) for $G = \mathbb{R}_{>0}$. $\square$ $\square$

COROLLARY 12.4 ([C]). (Phase Synchronization and the Reversible-Clock Holonomy.) [Conditional on Thm. 12.2; Thm. 11.42 on the phase horn; and local consistency of phase alignment — the spectral-difference data $\lambda_i - \lambda_j$ agreeing on overlaps, which is the cocycle condition for the phase transitions.] Take G the phase torus T^r (the structure group of the phase coordinate, r the frequency-module rank) and transition elements the relative phase alignments on overlapping orbits. Then phase coordinates synchronize across the family iff the phase holonomy vanishes on every independent

*cycle of the interface nerve — the obstruction in $H^1(N; T^r)$, supported on $\beta_1(N)$. This conditionally discharges *O_Phase.Sync*: synchronization is characterized with the holonomy named; its vanishing is branch data. In particular the reversible clock globalizes on tree-like orbit families ($\beta_1 = 0$) and is obstructed exactly by the cycle structure of the interface nerve.*

PROOF. On the phase horn the per-orbit coordinate is canonical up to origin-and-orientation (Thm. 11.42(II)), which is the $G = T^r$ gauge; relative alignments on overlaps are the transition elements, a cocycle by the assumed local consistency. Apply Thm. 12.2. $\square$

REMARK 12.5 ([R]). (The Obstruction Vanishes Exactly Where the Gauge Structure Is Trivial.) Thm. 12.2(2) places the synchronization obstruction on $\beta_1(N)$. The canon's cycle-rank theorem (**thm:beta1-controls-nonabelian**, § 13) places non-abelianity on the same β_1 : $\beta_1 = 0$ is the abelian sector, and β_1 is the rank of the non-abelian structure. These are the same support. The reading anticipated at the reversible-clock pass (**rmk:vrp-phase-connection**) is thereby a theorem-level identification: the failure of recovered variables to synchronize is the holonomy of a connection on the interface nerve, the connection is flat (consistent) by the cocycle condition, its holonomy lives on the cycles, and gauge structure is precisely the bookkeeping of that holonomy. Synchronization is free exactly when there is no gauge structure to obstruct it. Recorded as the program's reading of its own frontier; no force conferred beyond the cited theorems.

13. β_1 Cycle Rank as Canonical Complexity Parameter

DEFINITION 13.1 ([D]). (Cycle Rank β_1 of the LS-2 Adjacency Graph.) Let Γ_{LS2} be the LS-2 adjacency graph on the stabilized collar type alphabet Σ_∂ (Book II): the graph whose vertices are collar type classes and whose edges are admissible single-step collar transitions. The *cycle rank* (first Betti number) is

$$\beta_1(\Gamma_{\text{LS2}}) := |\text{Edges}| - |\text{Vertices}| + (\text{connected components}).$$

THEOREM 13.2 ([C]). (β_1 Controls Non-Abelianity.) [Conditional on $K_0 = 7$, LS-2, Phase-A.] *The cycle rank β_1 of the LS-2 adjacency graph determines the complexity of the non-abelian collar-local algebra:*

- (1) $\dim \mathfrak{g}' \leq \beta_1$ (Loop Rank bound);
- (2) $\beta_1 = 0$: *abelian sector only*;
- (3) $\beta_1 = 1$: *non-abelian sector contains $A_1 = \mathfrak{su}(2)$ ($|\Phi| = 2$)*;
- (4) $\beta_1 = 2$: *non-abelian sector contains $A_2 = \mathfrak{su}(3)$ ($|\Phi| = 6$)*.

At $K_0 = 7$: the color sector (B_{+1} block) has $\beta_1 = 2$; the weak sector (reduced active) has $\beta_1 = 1$; the background (B_0) has $\beta_1 = 0$.

PROOF. The dimension bound $\dim \mathfrak{g}' \leq \beta_1$ follows from the Collar-Local Lie Theory chain (SM Branch, Thm. ??, Step 5: Loop Rank). The classification by β_1 is the rank classification of compact semisimple Lie algebras: rank = β_1 selects the unique compact simple type with $|\Phi| \leq \rho = 8$. For $K_0 = 7$: the d_{S_v} -split gives three blocks with cycle ranks 2, 1, 0 respectively (from the NF2 class space structure, Book II, Def. ??). $\square$ $\square$

REMARK 13.3 ([R]). β_1 serves as a canonical complexity measure throughout the NFC program:

- In the YM branch: $\beta_1 = 2$ for the color sector drives the A_2 gauge algebra;
- In the NS branch: β_1 of the NS obstacle adjacency graph measures the difficulty of the regularity problem (higher β_1 = more independent obstruction modes);
- In the SM super-branch: combined $\beta_1^{\text{total}} = 3$ saturates the $\rho = 8$ obstruction bound ($|\Phi(A_2)| + |\Phi(A_1)| = 6 + 2 = 8$).

14. Boundary of Book III

What Book III proves.

- (1) Persistent observables are well-defined, quotient-visible, and transfer-compatible.
- (2) Admissible transfer maps compose, are witness-preserving, and respect the observational quotient.
- (3) The interface-relative stability functional satisfies the Unified Coercive-Transport Inequality at every admissible enlargement step.
- (4) Finite balance data compose along composable transfer chains.
- (5) Under extra coercivity hypotheses, observational drift is geometrically controlled.
- (6) Observational-transfer objects form a well-defined comparison arena.
- (7) Depth-sum convergence for Van Hove exhaustions (Thm 6.6): weighted boundary contributions vanish in the thermodynamic limit.
- (8) The ICDC Schema is recorded as an architectural scholium (Sch. 9.2): the abstract UCTI scale-limit formalization.
- (9) Two topological conservation scholia (Schs. 9.4–9.5): discrete-time Q_{simp} conservation and Hopf helicity conservation under admissible kernel evolution.

What Book III does not prove, and may not be assumed here.

- (1) *Universal source structure.* The existence of a universal source object $U \in \text{POT}$ is a theorem of Book IV. Book III builds the transport machinery that makes source completion meaningful; it does not prove source completion itself.
- (2) *Branch specializations.* The Yang–Mills specialization ($\Theta_n = 1$, gap-subcritical closure), the Navier–Stokes specialization ($\Theta_n = \theta < 1$, quasi-decay), and the gravity specialization all belong in derived branch books. No specialization of the Unified Coercive-Transport Inequality to a specific endpoint is canonical in this book.
- (3) *Continuum structure.* No continuum, differential, or smooth language is licensed at this stage. The transport inequality is a finite discrete bound.
- (4) *Branch closure.* Transport coercivity at this level is part of source-preparation discipline. Whether it extends to closure of any specific branch requires branch-level verification.
- (5) *Physical endpoint identification.* No identification with gauge fields, fluid dynamics, or geometric structure is claimed.

15. Handoff to Book IV

Book IV begins when the question shifts from *how local certified structure transfers* to *whether all such transfer data can be assembled into a single universal object*. The observational-transfer arena of Book III — objects with stabilized quotients, lawful transfer maps, and defect ledgers — is precisely the POT fiber data that Book IV will assemble into the universal completion.

The transition from Book III to Book IV is:

$$\begin{array}{lcl} \text{local admissible observables} & \longrightarrow & \text{stabilized persistent structure} \longrightarrow \\ \text{lawful transfer across contexts} & \longrightarrow & \text{universal source completion.} \end{array}$$

Book III supplies the first three terms. Book IV supplies the fourth.

16. Dependency Ledger

Theorem	St.	Depends on	Exports to
3.1 Persist. obs. quotient-vis.	[U]	SR 1.1, Def. 2.1	4.1, 9.1
3.2 Transfer preserves compat.	[U]	Def. 2.2	4.2
3.6 Composition well-def.	[U]	Def. 2.9	8.2, 8.4
3.5 Obstructions quot.-vis.	[U]	Defs. 2.1–2.7	4.4, 7.5
4.5 Compat. comparison	[U]	3.6, Def. 2.6	4.6, 9.1
4.6 Source-ready compat.	[U]	4.5	Book IV POT construction
6.2 Unified Coercive-Transport Ineq.	[U]	Books I–II, Defs. 5.3–5.7	6.5, 9.1; all branch books
6.5 Composable balance	[U]	6.2, Bk. I ledger additivity	7.3, 9.1
7.3 Finite Transfer Coercive Ineq.	[C]	6.5, coercivity hypotheses	Branch books (NS, YM, GR imports)
7.5 Obstruction blocks source	[U]	4.4, 4.6	Book IV; branch books
8.4 OT comparison arena	[U]	8.1, 8.2	Book IV POT; 9.1
9.1 Obs. Transfer Thm. of Spine	[U]	All of Bk. III	Books IV–VII; all branch books
6.6 Depth-Sum Convergence	[U]	Van Hove geometry; Bk. I finiteness	NS Branch Book (weighted boundary bounds)
9.2 ICDC Schema (abstract)	[O]	6.2; five conditions	Branch books Part III; YM/NS/GR/SCC ICDC
9.4 Topological conservation	[C]	Gate structure G1–G3	GR Branch Book remark
9.5 Hopf helicity	[C]	Admissibility of Φ	GR Branch Book remark

SCHOLIUM 16.1 ([R]). The Unified Coercive-Transport Inequality (Theorem 6.2) is the load-bearing core of Book III and is unconditional: it depends only on the defect

ledger of Book I and the LS-2 boundary law of Book II. The Finite Transfer Coercive Inequality (Theorem 7.3) is conditional, requiring extra coercivity hypotheses. This conditionality should remain explicit in every downstream citation. No branch book may promote the conditional result to unconditional force without discharging the extra hypotheses within the branch.